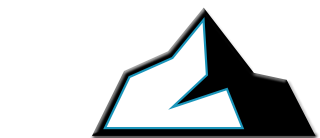
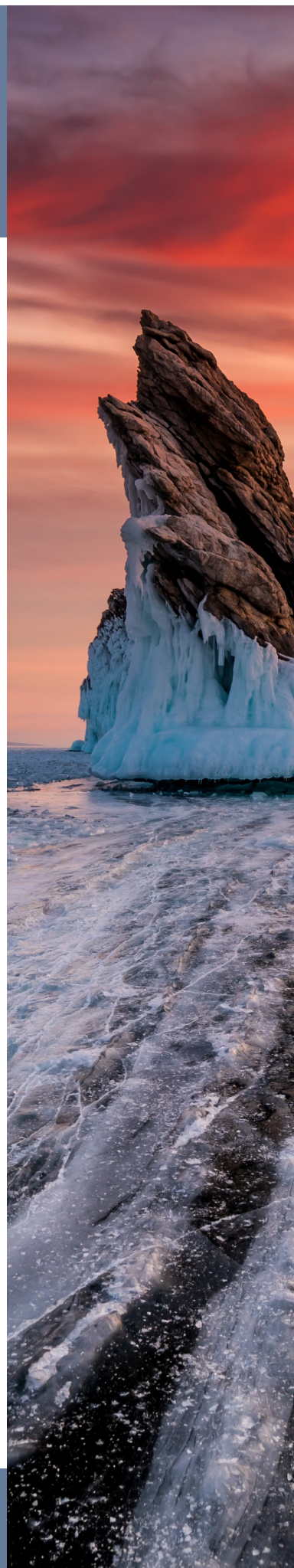


FIRE | FOSSA PP AURORA



ICELANDIC
WELLNESS

OWNER'S MANUAL



THANK YOU FOR JOINING THE ICELANDIC WELLNESS FAMILY, WE APPRECIATE YOUR BUSINESS!

Our design, engineering, and production team have ensured that only the finest quality materials and craftsmanship have been combined to provide you with years of enjoyment!

Please take a few minutes to read this owner's manual. It contains important SAFETY, installation, operating, and maintenance instructions.

For your home records, please fill in the purchase & delivery information. After receiving your tub, place the serial # sticker in the box below the form.

Date Purchased	_____
Date Installed	_____
Dealer	_____
Address	_____
Telephone	_____

Box for serial #/model

Scan to Register
Your Warranty



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SAFETY INSTRUCTIONS



WARNING! IMPORTANT SAFETY INSTRUCTIONS READ AND FOLLOW ALL INSTRUCTIONS

1. Avoiding Risks to Children

- Extreme caution must be exercised to prevent unauthorized access by children.
- Be sure the hot tub cover is closed and securely locked when not in use. Every Celtic Hot Tub is equipped with a locking cover that meets the ASTM F134691 standard for safety covers. Keep cover keys out of reach of children.
- Never allow children to use the hot tub without adult supervision.

2. Avoiding Risk of Electrocution

- Do not permit electrical appliances within 5 feet of the hot tub.
- Test the GFCI prior to every use of the hot tub.
- Do not use the hot tub with the equipment area door open.
- Do not use an extension cord to connect the hot tub to its power source.
- Disconnect the hot tub from its power source before draining the spa or servicing electrical components.
- Do not open the electrical control box. There are no user serviceable parts inside.

3. Increased Side Effects of Medication

- DRUGS AND ALCOHOL CAN GREATLY INCREASE HEALTH RISKS AND HYPOTHERMIA IN A HOT TUB.
- Persons with medical conditions or taking medication should consult their doctor prior to using a hot tub. Some medications may cause drowsiness, or affect blood pressure, heart rate or circulation.

4. Reduce Risk of Injury

- Do not use the hot tub immediately after a strenuous workout.
- Extended immersion in a hot tub could be injurious to your health.
- Pregnant or possibly pregnant women should always consult their doctor before using a hot tub.
- Never allow anyone to dive or jump into a hot tub.
- Always enter and exit your hot tub feet first, slowly and carefully. Floors and surfaces can be slippery.
- People with medical conditions such as heart disease, high or low blood pressure, circulatory system issues, obesity, or pregnant women should consult a doctor prior to using a hot tub.
- DO NOT Remove suction or filter covers. Water suction through the drains and skimmer is very strong when pumps are operating at high speed. Damaged or missing drain and skimmer covers can be dangerous entrapping small children and people as well as long hair. Should anyone or anything become entrapped in a drain, skimmer or any other fitting, **IMMEDIATELY TURN OFF THE HOT TUB AT THE ELECTRICAL SOURCE.**
- People with infectious diseases or sores should consult their doctor prior to using a hot tub.
- Always install cartridge filters prior to filling the hot tub. If you are changing a filter when the hot tub is full of water, be sure to turn off the hot tub before removing the dirty cartridge filter and installing the clean cartridge filter.



INTRODUCTION

Welcome to your new spa. With the correct preparation and care, your new spa will provide you with many years of fun and relaxation. Please find the "User Manual" and "Quick Start Guide" in the product attachment.

This manual has been developed to provide you with the information that you need regarding the preparation, installation, care and operation of your new spa. Please take the time to carefully read the entire manual to ensure that your spa preparations are carried out correctly. Also make sure that you familiarize yourself with the important safety instructions before using your spa.

If you have any question or doubt on the operation or maintenance of this product, please contact your local dealer.

MAIN FUNCTION AND PRODUCT CHARACTERISTICS

HYDROTHERAPY FUNCTION: The product comes with special hydrotherapy seats which are designed according to ergonomic engineering, different kinds of jets in the seat to help to achieve body therapy. Adjust the massage pressure by adjusting the face, water diverter and air regulator.

AIR BLOWER THERAPY FUNCTION: Turn on the air blower, specially designed air jets will blow lots of air bubbles to provide you bubble therapy and fun.

HEATING FUNCTION: SPA comes with heater, once the user set the temperature, the heater will start to maintain the temperature to the set temperature and ensure the spa is available for all sea-son.

FILTER AND CIRCULATION FUNCTION: SPA comes with Filter which can clean the impurities inside efficiently, meanwhile, the system will run the Filter Cycle programming automatically, to kill the bacteria via the ozone released by the ozone generator.

AROMATHERAPY FUNCTION: This spa is fitted with an aromatherapy canister which contains scented oil. The soothing scent is dispersed into the water through special plumbing to help you relax.

LIGHTING: The spa is fitted with an underwater light. The color and mode of the light can be changed via the control system touch panel, helping you create an ambience to suit your mood.

MULTIMEDIA: The optional DVD/TV/IPOD allows you to enjoy video or music while exercising in the swim spa or relaxing in the spa.

SWIMMING (SWIM SPA): Adjust the swim jets to create a powerful current which allows you to enjoy endless swimming without taking up a lot of space.

SURF SPORT (SWIM SPA): Adjust the direction of swim jets to create a vortex. Enjoy the effect of surf in your swim spa.

EXERCISING (SWIM SPA): With the help of an exercise band or swimming harness, you are able to do all kinds of exercise in swim spa.

Important declaration: All functions mentioned above are a general introduction to our spas products, The SPA you purchase may be different. Please check it thoroughly before purchasing. The manufacturer reserves the right to modify and improve the product when necessary without prior notice and without entering into any obligation.

IMPORTANT INFORMATION

It is essential to familiarize yourself with general spa pool safety and water treatment procedures. Please make sure to familiarize below contents:

DRAINAGE REQUIREMENT:

Ensure that the spa is not in a low or floodable position; as this could cause electrical equipment short-age or be dangerous to spa users

RISK OF ELECTRIC SHOCK: As with all electrical appliances there is a risk of electrocution. The engine room within the cabinet, houses the control unit that operates the spa. The door enclosing the unit should always remain shut. If water happens to enter the engine room, turn the spa off at the power outlet and make sure all the components are dry before attempting to power the spa back up.

TEMPERATURE: Keep your spa water temperature no higher than 40°. Higher water temperatures can cause strain on the cardiovascular system. If the water temperature is higher than 40°, get out of the spa and contact the dealer as soon as possible. It is recommended that the water temperature be no higher than body temperature i.e. 37.5°.

MEDICAL CONDITIONS

- People with diabetes, high blood pressure, heart disease or other cardiovascular conditions should consult their physician before using a SPA.
- To reduce the risk of contaminating spa water and minimize health risks, shower with soap and water prior to using the spa.
- People with open sores or any type of infection should not use the spa.
- Hot water is an ideal environment for spreading infection, especially-if sanitizer levels are not properly maintained.
- Women who are pregnant should consult their physician before Using a spa.

SOAK AND COOL TIME

Limit spa use to no more than 15-20 minutes at a time. Some spa pools are equipped with a cooling seat. When you feel you are getting too hot, sit on the cooling seat and allow your body to cool. When you feel comfortable you may submerge yourself back into the spa pool. Sitting in the spa with your heart above the water level will dramatically decrease-the time it takes to cool down. Hypothermia may be fatal. Do not expose your body to prolonged exposure in water above normal body temperature.

ALCOHOL USE AND MEDICATION: The consumption of alcohol in and around the spa pool is a hazardous practice. Never use glass drinking vessels near the spa pool. Alcohol can cause drowsiness-and lead to drowning. Any person taking medication should consult their physician before entering the spa pool.

CHILD SAFETY: Children should not be permitted to use the spa without adult supervision.

PREVENTING SLIPS: Always take care when entering and exit-ing a spa due to slippery, uneven surfaces. Handrails and non-slip surfaces will help to minimize the risk of slipping and falling. In some cases steps may be needed to make entering the spa easier.

LOCATION REQUIREMENT

Before installing, it is better to ask a professional to check the spa position. Avoid placing the SPA where water could flood into the electrical equipment compartment, and also protect the unit from direct contact with very damp areas.

IMPORTANT INFORMATION

ELECTRICAL APPLIANCES: Never have any electrical devices within 2 meters of the spa.

WEATHER CONDITIONS: Do not operate or soak in a SPA during severe weather conditions i.e. thunderstorms, cyclones or electrical-storms.

VENTILATION: If the SPA pool is not sufficiently ventilated some people may experience throat irritations. Make sure that the SPA pool is properly ventilated if you are installing it in an enclosed area.

CHEMICALS: SPA chemicals are necessary to control bacteria, prevent infection and maximize the life of your SPA and its equipment. However, in concentrated form these chemicals, like other household chemicals, can be hazardous. Always handle and use them with care. Refer to the product's label for safety and handling information. A more in-depth explanation on how to use chemicals in your SPA can be found in the Water management & maintenance.

SPA INSTALLATION

Please always find a professional to install your SPA, The manufacturer shall not be responsible for any damage caused by an improper installation or an installation performed by non-qualified personnel.

POSITIONING AND LOCATION OF THE SPA

- Your spa needs to be positioned so that the spa's equipment is easily accessible. The equipment is located (in most cases) at the front of the SPA and will need to be accessible for service and power connection. It is recommended that you have access to all sides allowing at least 1/2 a meter access around all 4 sides of your SPA for future access if required. It will be the owners responsibility to make sure this is completed prior to the service being carried out. All portable SPA equipment must be ventilated. Do not obstruct any vents in the cabinet.
- It is recommended to place the SPA away from trees, as falling leaves may block the filter.



Important note: Do not leave your SPA uncovered in direct sunlight. The SPA has closed cell insulation foam under the shell. If left exposed to direct sunlight the surface temperature can build up to 200° causing damage to your SPA. It is suggested that you have the hard cover over your SPA while not in use. Leaving the SPA uncovered and exposed may void your warranty on your SPA and its accessories-.

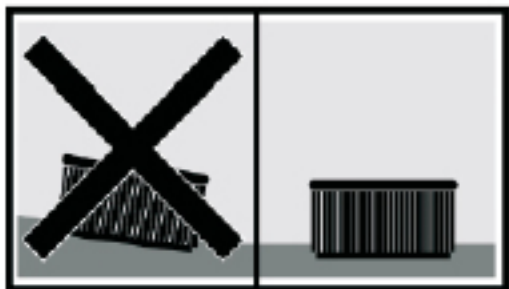
SPA INSTALLATION

**** IMPORTANT ****

- Position the SPA horizontally, placing the whole base on a smooth, flat and level surface capable of supporting the weight when used (The weight of the SPA, the water and the weight of the bathers need to be taken into account)
- The SPA cannot be placed on a curved, soft, sandy surface or blocks.

LOCATION REQUIREMENT

- Before installing, it is better to ask a professional to check the spa position.
- Avoid placing the SPA where water could flood into the electrical equipment compartment, and also protect the unit from direct contact with very damp areas.



ELECTRIC INSTALLATION

- The electrical supply to your new spa will require a dedicated circuit with no other appliances or sockets sharing the power. Electrical burnout due to incorrect amperage WILL NOT be covered by warranty. All electrical connections must be carried out by a qualified electrician.
- Ensure your new SPA is safely earthing. Ensure the main power supply has installed high sensitive leakage protectors. Recommend 30mA leakage protectors.
- Refer to your specific spa specs for various electrical needs

SPA MAINTENANCE

Owning a spa is fun, relaxing and therapeutic, but more importantly we must look after our investment to keep it healthy and economical, thus avoiding unnecessary premature repair costs. Detailed below is a simple step-by-step procedure to follow.

SPA INSTALLATION

FILLING YOUR SPA

- Before filling your spa, please check and tighten all barrel unions
- on pumps and heater as they may become loose during transit.
- Please check and make sure all the check valves are all in the open position.
- Check the drainage valve is shut off.
- Check that all jets are in the open position. Turn anti-clockwise to turn your jets on.
- Fill your SPA with water. Fill the water through the filter in order to
- Prevent air blockages in the pumps.
- Fill water to the line on the water level indicator.

The required water level



Note: To avoid any damage to the components, do not fill hot water over 104° into your spa

DRAINING YOUR SPA

- If you use your SPA once or twice a week, the need to drain your spa will be less frequent than a SPA owner who uses their SPA three or more times a week. Therefore, it would be recommended that a SPA be drained at least every 3 months.
- Your drainage valve is positioned at the front corner of the base. Details for drainage are as follows:



1. Remove the cap of the drain valve by turning anti-clockwise.



2. Connect a 3/4 inch hose or screw a standard male hose fitting onto the valve and connect your hose.



3. Turn value anti-clockwise.



4. Pull outwards and the SPA will begin draining.

Note: Ensure the power is shut off before draining. Equipment damage caused by dry running will not be covered under warranty

SPA CONTROL PANEL

- For SPA control panel operation, please refer to the User Manual of the Control System.

JET INSTRUCTION

- Jets are the main parts in which massage comes into effect.
- Turn the jet face clockwise about 1/4 of a turn, the jets are turned on.
- Turn the jet face anti-clockwise about 1/4 of a turn, the jets are turned off



Note: Some jets and the air jets are unadjustable. Don't rotate these jets with force. It may cause damages to the jets.

AIR CONTROLS AND WATER DIVERTER

AIR CONTROLS INSTRUCTION

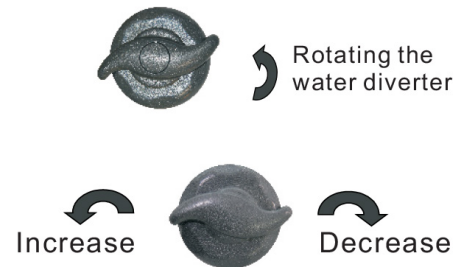
- Air controls affect the pressure of your jets by adding air. You may have more than one air control on your spa, each of which will be associated to a certain group of jets. By opening an air control when the pump is on air will be drawn into the water flow.
- When the air control is turned clockwise, it will decrease the power of the massage jets.
- When it is turned anti-clockwise, it will increase the power of the massage jets.



Note: For better heating efficiency, please turn the air controls off. The air may cool the spa water temperature.

WATER DIVERTER

- The Water Diverter allows you to adjust the massage power. Rotating the water diverter will divert the massage power of different seats.



The Water Flow Control controls the flow of water to the options, such as waterfalls and fountains. Rotating the water flow control will turn on, turn off and adjust the water flow.

Note: For better heating efficiency, turn the waterfalls off when the spa is not in use.

SPA WATER MANAGEMENT & MAINTENANCE

- Please note that these instructions are just a guide to maintain the spa's water quality. Before any actions, please consult your local Spa specialist for proper amount of sanitizer use, and take proper action depending on the capacity of the spa.
- You should never use pool grade chemicals in your spa. Any damage to equipment caused by using pool grade chemicals or other improper chemicals will not be covered under warranty.
- Spa water should be professionally tested at least once a month, it's recommended that the water is tested weekly by the end user.
- When adding chemicals to the spa, make sure the pumps are running and allow 20-30 minutes for the chemicals to mix. Test the water after the chemicals are sufficiently mixed into the water.

Note: Do not cover the SPA for at least 30min after adding chemicals.

THE EFFECT AND ADJUSTMENT OF PH

- Low pH (below 7.2) causes sanitizers to dissipate more rapidly increases corrosion of surfaces and equipment, and may make water irritating to spa users. If pH is high (above 7.6) chlorinating sanitizers are less effective, surfaces and equipment can scale, water may cloud and shorter filters runs may occur.

ATTENTION: The pH level measures the acidity and alkalinity: Values above 7 are alkaline and below 7 are acid. A Ph index between 7.2 and 7.6 is recommended

WARNING: Please make sure that the correct PH level is maintained, any damage caused by an inadequate PH level is not covered by your Spa guarantee.

PH ADJUSTMENT

- Check the pH of the Spa water daily using the pH tester. (not supplied)
- If the pH is above the above-mentioned indexes, do not enter the SPA temporarily. Wait for two hours and re-do the pH test.
- If the pH is below the above-mentioned indexes, let the pumps prime. Wait for two hours and re-do the pH test.
- If the proper pH level can not be maintained by the above instructions, please add appropriate amount of acidity-alkalinity balance reagent into the water, uncover the spa cover, leave it mix in for 20 minutes, then re-do the pH test.

DISINFECTION OF THE WATER

- Disinfection of the water is of utmost importance in order to destroy algae, bacteria and organisms that could develop in the water. However, excessive disinfection could cause skin and eye irritation.
- **Bromide tablets:** Are a suitable disinfectant for the Spa water. This product is placed in the pre-filter and gradually dissolves. Check the residual bromide level daily using the Br analyser set. The recommended level of residual bromide is between 2.2 and 3.3 ppm.
- **Disinfection with OZONE:** Ozone or O₃, is an oxidizing chemical, which is very effective for disinfecting water. The main advantage of ozone is that it leaves no chemical trace and is odorless. Generally the spa is equipped with 1 ozone generator which produces ions from oxygen in the atmosphere. This is an automatic process and resulting product is injected into the water through the ozone jets. It is recommended that the ozone disinfection is operated twice a day and the duration is maintained within 1~2 hours, thus the ideal disinfection effect can be obtained and the life span of the ozone generator can be extended.

OWNER'S MANUAL

THE APPLICATION OF OTHER SPECIAL CHEMICAL PRODUCTS

- Besides the pH adjustment and the above-mentioned disinfection ways, there are some cleaning products which can be used in spa water;
- ANTI-CALCAREOUS: it can be used to prevent/reduce precipitation of calcium salts (scale), especially in hard water areas. Use it once a week and whenever the water is renewed.
- SPA ALGAECIDE: Prevent algae from growing in the water, use it once a week and whenever the water is renewed.
- ANTI FOAM: Prevent precipitation of calcium salts (scale), use it once a week and whenever the water is renewed.

WARNING!

Add the exact amounts to the water, as specified. Do not mix products, to avoid possible reactions.

Do not smoke while handling these products – they may be flammable.

Do not add chemical products to the water if there is someone in the spa.

Keep containers tightly closed in dry, well-ventilated places, and keep these products away from children.

Do not inhale chemical products, and take care not to let them come into contact with the eyes, nose or mouth. Wash hands after use.

Follow the emergency instructions on the product label in the event of an accident or ingestion.

SPA CLEAN AND MAINTENANCE

The SPA surface is made of high gloss acrylic. It is recommended that the spa surface (above the water line) be cleaned once a week to remove dirt build up and scum lines. The whole spa surface should be cleaned when the spa is drained. Cleaning details are below.

SPA SURFACE CLEAN AND MAINTENANCE

- The spa surface is made of high-gloss acrylic, we must keep it clean, we suggest that clean the spa outside surface once a week and clean the whole spa after drain. Details steps as below:
- Clean the surface with wet towel. A neutral detergent and soft towel may be used. Do not use detergent with ketone or chlorine water.
- If the surface has some light scratches, it can be polished with a fine cutting compound and finished off with a fine polish.
- Please do not clean the spa surface with any chemical solution/Kernel or crude tools; do not scratch the spa surface with hard
- Do not touch the spa surface with any heat source over 70° material or knife.
- Please do not clean the spa surface with nail polish, nail polish remover, dry detergent, acetone, paint remover ect. It will damage the surface.

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CONTROL PANEL MAINTENANCE

- Please do not touch the control panel with hard objects to avoid scratching.
- Please avoid prolonged direct sunshine on the control panel.
- Please keep the control panel covered when not in use.

HEADREST MAINTENANCE

- Clean the headrests with a neutral detergent.
- Excessive chemical levels and ozone may cause fading of headrests. Headrests can be removed after use. Fading is not covered by warranty.

SPA COVER MAINTENANCE

- Clean the spa cover with soapy water.
- Do not pull or carry the cover by the link in the middle of the cover.
- Do not sit, lie or jump on the cover, damage this may cause is not covered by warranty.

NOTE:

Please cover the spa cover after use or clean. Leaving the spa uncovered and exposed may void your warranty on your spa and its accessories.

FILTER MAINTENANCE

- You should clean your filters regularly. Once a week you should remove your filters and hose any debris out of them with a high pressure hose. Once a month you should soak your filters in a cartridge filter cleaning solution and then hose off with a high pressure hose.

FILTER CLEANING

- Turn off the power
- Remove the cartridge (please read the filter removal instruction before doing so).
- Remove the cartridge by turning counter clockwise.
- Clean the cartridge with water to remove debris by spraying from top to bottom at a 45° angle.
- Install the filter and cartridge after cleaning.
- Turn on the power.



WARNING:

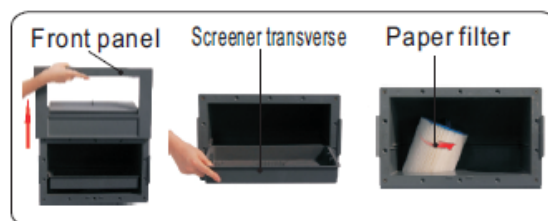
Before removing your filters, ALWAYS ensure that the power to your spa is turned off. If you leave your power on and anything is drawn into the pump(s) or heater, causing either to fail, the fault will not be covered under warranty.

OWNER'S MANUAL

FILTER REMOVE AND INSTALL

REMOVE THE CMP CARTRIDGE AND FILTER

- Open the cartridge cover.
- Turn the cartridge counter clockwise to remove.
- Remove the screener over the filter.
- Remove the filter.
- Go the opposite way to replace the filter and cartridge.



REMOVAL OF TWIN FRONT ACCESS FILTERS

- Lift the front panel upwards to remove and pull screener out.
- Turn the filters counter clockwise to remove.
- Go the opposite way to replace filters.



REMOVE BIG SIZE FRONT ACCESS STOPPER FILTER

- Take off the top cover.
- Go counter-clockwise to remove the paper filter.
- Go opposite to install the front access stopper filter.

WHEN SPA IS NOT IN USE

If you are going away for a period longer than 2 months, it is recommended that you turn your spa off and empty it. It is important to remember to leave the cover on even while empty to protect the shell. If you are going away for a few days or weeks, do not turn your spa off completely. Turn the temperature down to 15 degree, so that the filtration cycles are still functioning but the spa doesn't heat.

3-5 days

Test the spa water to check PH and sanitiser levels and adjust if necessary. Cover the spa. Test water balance and adjust if necessary before using the spa again.

5-14 days

Set the spa to the lowest temperature setting. Check the water balance and adjust if necessary. Cover the spa. Check water balance and adjust if necessary before using the spa again.

More than 15 days

Turn the power off, open the drain valve and drain the spa clean, remove the filter, dry the spa and cover with the spa cover.

BP100G1 Tech Sheet

Customer: Balboa Water Group

Part Number: 59187-01 5.5kW 800 Incoloy
59188-01 4.0kW 800 Incoloy
59251-01 Remote Heater System -- Heater is sold separately

Custom Box Overlay ☐

Box Overlay Part Number N/A

UL System Model For 5.5kW: BP1-BP100G1-BU

UL System Model For 4.0kW: BP1-BP100G1-BS

UL System Model For Remote: BP1-BP100G1-B

Software Version ID: M100_230 V52.0

Software Version: 52.0

File Name: BP100_52.0_BP100G1.hex

Configuration Signature: E1177576

Eng. Project Number: 5270

Control Panels (See later pages for more information):

spaTouch™2	Any version (version 2.0 or later required for bba™2 fully integrated functionality)
Icon spaTouch™	Any version (version 3.36 or later required for bba™2 fully integrated functionality)
Menued spaTouch™	Any version (version 2.8 or later required for bba™2 integrated functionality)
TP900	Version 3.1 and later (Version 3.13 or later required for bba™)
TP800	Version 3.1 and later (Version 3.13 or later required for bba™; version 4.11 or later required for bba™2 integrated functionality)
TP600	Version 2.7 and later (Version 2.12 or later required for bba™/bba™2 On/Off control via menu)
TP500	Any version
TP400T US	Version 2.7 and later (TP400T CE may be used) (Version 2.12 or later required for bba™/bba™2 On/Off control via menu)
TP400W US	Version 2.7 and later (TP400W CE may be used) (Version 2.12 or later required for bba™/bba™2 On/Off control via menu)

Basic Functions Setup 1 - 3

Power Requirements:

240VAC, 50/60Hz*, 48A, Class A GFCI-protected service (Circuit Breaker = 60A max.),
4 wires [hot, hot, neutral, ground]

120/240VAC, 50/60Hz**, 16/40A, Class A GFCI-protected service (Circuit Breaker = 20/50A max.)
3 or 4 wires [hot, hot (optional), neutral, ground].

****NOTE:**

The above 120V spec is only when using a wall-mount GFCI / breaker.
If using a GFCI cord, the breaker is 15A and so the service is limited to 12A.

*BP systems automatically detect 50Hz vs 60Hz. However, power frequency (50Hz vs 60Hz) is just one of many differences between North American (UL) and CE power, and it is because of these other differences that different BP systems must be used for UL vs CE territories. Also, there are a few countries that use CE power but 60 Hz (such as South Korea) which need CE systems, and a few countries that use UL power but 50 Hz which need UL systems.

HiPot Testing Note:

Disconnect slip terminal with green wires from J6 prior to performing HiPot test. Failure to disconnect may cause a false failure of the test. Reconnect terminal to J6 after successful completion of HiPot test.

Basic Functions Setup 1 - 3

System Outputs:

Pump 1	240VAC*	2-Speed	12A max	15-minute timer (30-minute timer for P1 Low in non-circ Setup 2 only)
		This is the heater pump in Setup 2. Must deliver 20 GPM through heater		
		1 Speed in Setup 1 Unused in Setup 3		
Circ Pump	240VAC*	1-Speed	2A max	Programmable Filtration Cycles + Polling
		This is the heater pump in Setups 1 & 3. Must deliver 20 GPM through heater		
Ozone	240VAC*		.5A max	Slaved to Circ Pump in Circ Setups and to Pump 1 Low in Non-Circ Setups
Spa Light	10VAC	On/Off	1A max	240-minute timer.
A/V (Stereo)	120VAC	Hot	4A max	Always on
Heater	5.5kW @ 240VAC (approx 1.4kW @ 120VAC)			
	4.0kW @ 240VAC (1.0kW @ 120VAC)			

*Pump 1, Circ Pump and Ozone must be the same voltage.

With 120VAC power input, Pump 1, Circ pump and Ozone must all be 120V. See wiring diagram for rewiring instructions.

Setup Reference Table

Setup #	Circ Pump	Pump 1	Temp Scale
1	Programmable Filtration + Polling	1-Speed	°F
2	None	2-Speed	°F
3	Programmable Filtration + Polling	None	°F

System (and any replacement board) **is shipped in Setup 2**

Changing Software Setups with spaTouch™ Icon-Driven Panels

Test Menu Access (S1, Switch 1 ON) *Service Technician ONLY.*

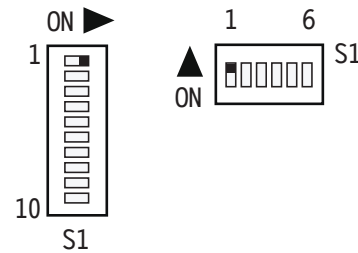
DANGER! HIGH VOLTAGE WILL BE ACCESSIBLE! SERVICE TECHNICIAN ONLY!

While the system is running, move DIP Switch 1 (on S1 on the Main circuit board) to ON. The system will enter Test Mode.

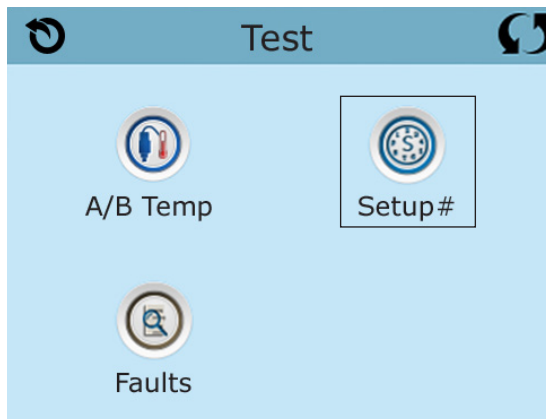
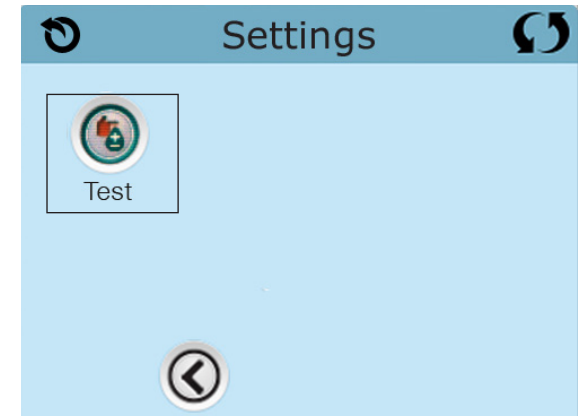
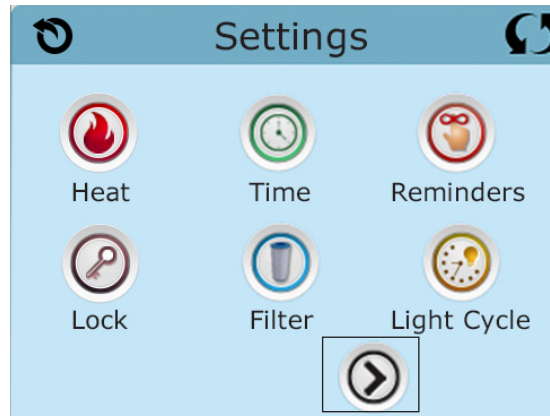
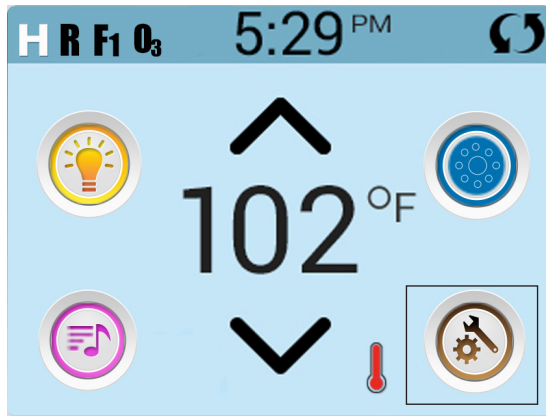
Moving DIP Switch 1 to OFF will exit Test Mode.

To Change Software Setups:

While in Test Mode, press the indicated icons to move from screen to screen.

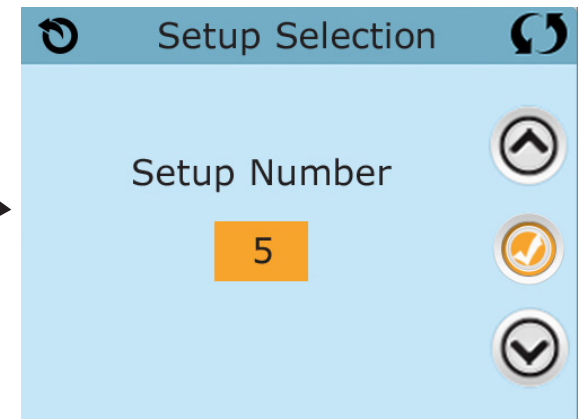


The example screens shown here are from the spaTouch 1 Icon-Driven Panel, but the screens on the spaTouch 2 Panel are similar. The main difference is that the spaTouch 2 display is wider.



Once on the Setup Selection screen, press the Up or Down icon to select the desired Setup Number, then press the Check Mark icon to confirm and to have the spa restart.

After the system restarts, you may see a message that "The settings have been reset"; this is normal after changing Setups with DIP Switch 6 in the OFF position. Press "Clear" to dismiss this message.



Changing Software Setups with TP800 / TP900 / spaTouch™ Menued Panel

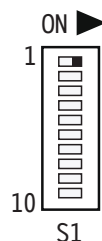
Test Menu Access (S1, Switch 1 ON) *Service Technician ONLY.*

DANGER! HIGH VOLTAGE WILL BE ACCESSIBLE! SERVICE TECHNICIAN ONLY!

While the system is running, move DIP Switch 1 (on S1 on the Main circuit board) to ON.

The system will enter Test Mode.

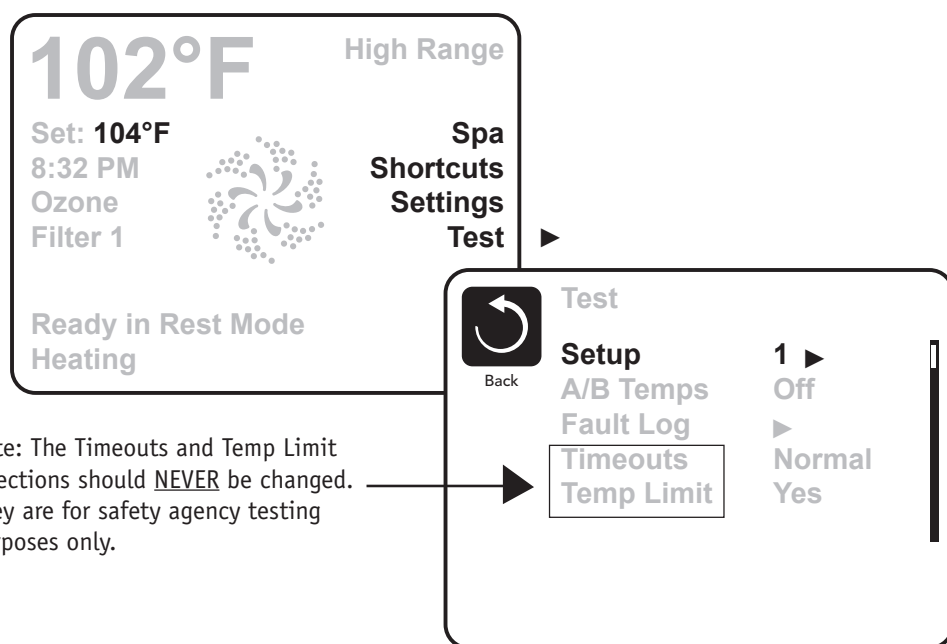
Moving DIP Switch 1 to OFF will exit Test Mode.



Software Setups

Under the TEST Menu, the Setup screen will allow changing the Setup from 1 to any number established by the Manufacturer.

Changing the Setup may require wiring changes as well.



Changing Software Setups with TP600 / TP500 / TP400

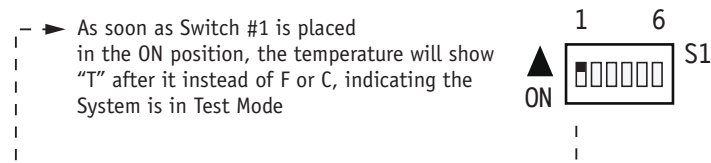
Test Menu Access (S1, Switch 1 ON) *Service Technician ONLY.*

DANGER! HIGH VOLTAGE WILL BE ACCESSIBLE! SERVICE TECHNICIAN ONLY!

While the system is running, move DIP Switch 1 (on S1 on the Main circuit board) to ON.

The system will enter Test Mode.

Moving DIP Switch 1 to OFF will exit Test Mode.



Software Setups

Under the TEST Menu, the Setup screen will allow changing the Setup from 1 to any number established by the Manufacturer.

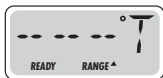
Changing the Setup may require wiring changes as well.

You will have 1 minute to complete the setup change after you manually exit Priming Mode. (Once familiar with the process, the Setup change should take less than 15 seconds.)



When the panel displays RUN PMPS PURG AIR, press any Temperature button ONCE to exit Priming Mode.

You should see "---T" where the T indicates the system is in Test Mode.



Continued on Next Page.

Changing Software Setups with TP600 / TP500 / TP400 Continued

Again, **You will have 1 minute** to complete the setup change after you manually exit Priming Mode.

NOTE: Wherever the below says Warm or Temp followed by Light, on the TP500 press Menu instead of Warm or Temp followed by light. And whenever the chart below says Light, on the TP500 press Menu instead of Light.

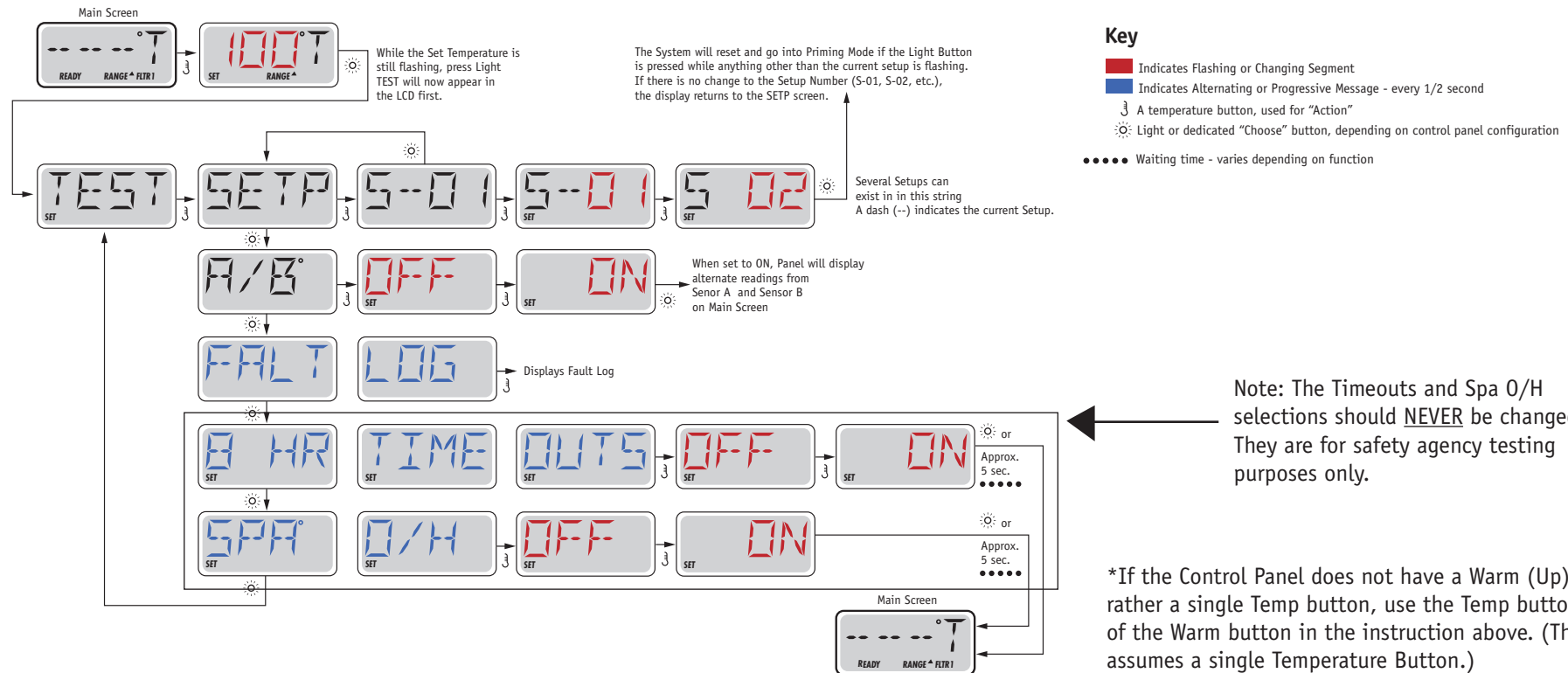
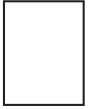
Immediately after exiting Priming Mode, press this sequence of buttons: Warm*, Light, Warm, Warm, Warm, Warm. Continue to press Warm until the display shows the Setup Number (S-01, S-02, etc.) you want to switch to. When the correct setup number is showing, press Light once, and the system will reset, using the newly-selected Setup from that point on.

Move DIP Switch 1 to the OFF position to take the spa out of Test Mode. °F or °C will replace °T.

Using a permanent marker, write the Setup number on the Setup label mounted inside the system lid (right). This is very important to any service person in the future who may need to replace a circuit board or system and needs to change the Setup on a replacement part while in the field.

NOTE: Changing the Setup may require wiring changes as well - refer to the wiring diagram or wiring diagram addendum.

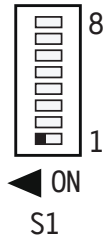
THIS SYSTEM IS
CONFIGURED AS
SETUP #



DIP Switch Functions

Fixed-fuction DIP Switches

- A1 Test Mode (normally Off).
- A2 In "ON" position, add one high-speed pump (or blower) with Heater.
- A3 In "ON" position, add two high-speed pumps (or 1 HS Pump and Blower) with Heater.
- A5 In "ON" position, enables Special Amperage Rule B. See Special Features section under Configuration Options for functionality with your system.
In "OFF" position, enables Special Amperage Rule A.
- A6 Persistent memory reset (Used when the spa is powering up to restore factory settings as determined by software configuration).



A2 and A3 work in combination to determine the number of high-speed devices and blowers that can run before the heater is disabled. i.e. A2 and A3 in the ON position will allow the heater to operate with up to 3 high-speed pumps (or two HS Pumps and Blower) running at the same time. Heat is disabled when the fourth high-speed pump or blower is turned on.

Note: A2/A3 all off = No heat with any high-speed pump or blower.

Assignable DIP Switches

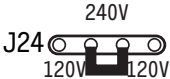
- A4 In "ON" position, enables a 5-minute cooldown for some gas heaters (Cooling Time B).
In "OFF" position, enables a 1-minute cooldown for electric heaters (Cooling Time A).
- A7 In "ON" position, Simplified Menus on TP400/TP500/TP600. Do not use graphic panels (TP800, TP900, or spaTouch™ family) with Simplified Menus.
In "OFF" position, Regular Menus on TP400/TP500/TP600. This setting is compatible with all panels.

Undesignated switches are not assigned a function.

WARNING:

Should only be done by an authorized service technician

Jumper Definitions

J109	GFCI Test/Trip Enable/Disable Note: This feature must be enabled in software as well.	J109 
J30	Do Not Use	
J31	Jumper on 1 pin when heater voltage is 120V Jumper on 2 pins when heater voltage is 240V	J31 
J24	Jumper on center two pins (240V) when heater voltage is 240V. Two Jumpers installed; one on left 2 pins and one on right 2 pins (120V) when heater voltage is 120V.	

Warning!

Setting DIP switches or jumpers incorrectly may cause abnormal system behavior and/or damage to system components.
Refer to Switchbank illustration on Wiring Configuration page for correct settings for this system.
Contact Balboa if you require additional configuration pages added to this tech sheet.

Replacement Parts

PCBA:

Main PCBA:	59189-01
Expander PCBA:	N/A

HEATER(s):

Heater:	58421 5.5kW 800Inc -- for integral heater only
	58426 4.0kW 800Inc -- for integral heater only
Temp Sensor Kit:	30344KIT 12-inch sensor -- for integral heater only
	30382KIT 24-inch sensor -- for integral heater only

FUSES:

Part Number	Amperage	Location
30136	30A	F6
26983	1A	F4, F5
24514	0.1A SLO	F3 when using 240V heater
26982	0.16A SLO	F3 when using 120V heater

BP100 Configuration Options

General Features

Feature	Default	
Pump 1 in Filter Cycle (Circ Only)	No	
Pump 1 Low Timer	30 Minutes	Applies in non-circ Setups (configurations) only
General Pump Timer	15 Minutes	
Blower Timer	15 Minutes	
Mister Timer	15 Minutes	
Light Timer	240 Minutes	
Circ (when enabled)	Programmable + Polling	
Cleanup Cycle	<i>30 Minutes</i>	
Cleaup as Preference setting	<i>Yes</i>	
Ozone	With Heater Pump*	
Ozone Suppression	OFF	
Pump Purge	60 Seconds	
Blower Purge	30 Seconds	
Mister Purge	5 Seconds	
Purge Type	Serial - Pumps at lowest speed	

* The heater Pump can be either a Circ Pump or Pump 1 Low.

BP100 Configuration Options

Temperature Features

Feature	Default																			
Temperature Display	°F																			
<i>All temperatures must be specified in °F. The system converts °F to °C dynamically. If Celsius is required for default settings, choose a desired °C value that (after rounding) corresponds to a Fahrenheit value.</i>																				
°C	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	
°F	39	41	43	45	46	48	50	52	54	55	57	59	61	63	64	66	68	70	72	
°C	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40		
°F	73	75	77	79	81	82	84	86	88	90	91	93	95	97	99	100	102	104		
Hi-Range Min. Set Temp	80°F																			
Hi-Range Max. Set Temp	104°F																			
Hi-Range Default Temp*	100°F																			
Lo-Range Min. Set Temp	50°F																			
Lo-Range Max. Set Temp	99°F																			
Lo-Range Default Temp*	70°F																			
Freeze Threshold	44°F																			
Freeze Type	Rotating - Pumps at Lowest Speed																			
Temp Lock Type	Temp + Settings																			

**May be changed by end-user (if enabled)*

BP100 Configuration Options

Time Features

Feature	Default
Time Format*	12 Hour
Filter 1 Start Hour*	20:00 (8:00 PM)
Filter 1 Duration*	2 Hours
Filter Cycle 2 Default*	OFF
Filter 2 Start Hour*	08:00 (8:00 AM)
Filter 2 Duration*	15 Minutes
Light Cycle	Disabled
Light Cycle Default*	OFF
Light Cycle Start Hour*	21:00 (9:00 PM)
Light Cycle Duration*	15 Minutes
Cooling Time A	1 Minute
Cooling Time B	5 Minutes

**May be changed by end-user (if enabled)*

BP100 Configuration Options

Reminder Features

Feature	Default
Reminders Shown*	<i>Yes</i>
Check pH	<i>OFF</i>
Check Sanitizer	<i>OFF</i>
Clean Filter	30 Days
Test GFCI	OFF
Drain Water	<i>100 Days</i>
Change Cartridge	OFF
Clean Cover	<i>OFF</i>
Treat Wood	<i>OFF</i>
Change Filter	365 Days

**May be changed by end-user (if enabled)*

BP100 Configuration Options

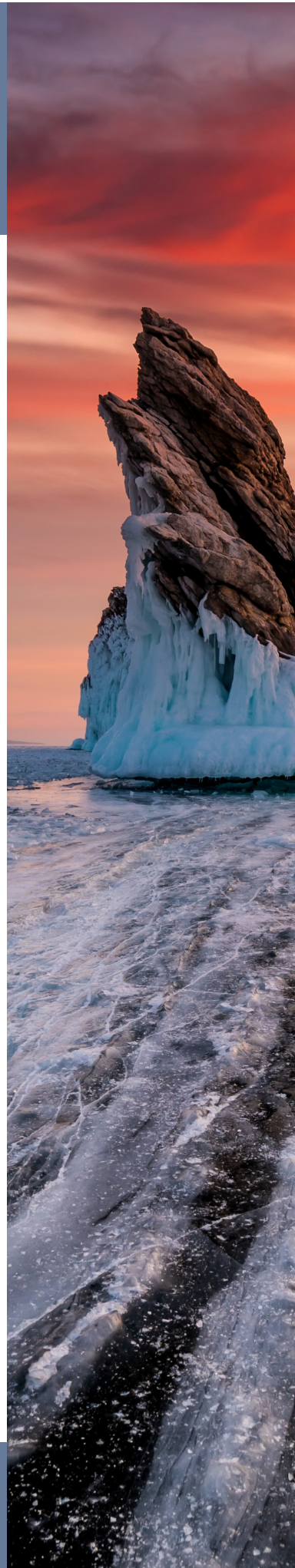
Special Features

Feature	Default
Special Amperage Rule A	No Limitation
Special Amperage Rule B	No Limitation
Drain Mode	Disabled
Demo Mode	Disabled
GFCI Trip	Enabled
Automatic GFCI Test	Disabled
Ozone Slaved to Heater Pump	Yes
Dual Voltage Heater	Always Input Voltage
Safety Suction	Disabled
Menu Style	Standard Menus when DIP switch A7 is OFF. Simplified Menus when DIP switch A7 is ON..

CONTROL PANEL

TP500S

Fire, Fosa PP and Aurora



TP500S Control Panels

User Guide for Standard Menu

System Model: All BP series systems

Panel Model: TP500S Series All

Panel Software Version: versions



Display Icons



A - Heat	F - Light	K - Auxiliary (Jets 3 or MICRO SILK [®])
B - Ready Mode	G - Cleanup Cycle	L - Temperature Range (High / Low)
C - Rest Mode	H - Jets 1	M - Set (Programming)
D - bba™2 On	I - Jets 2	N - Filter Cycle (1 or 2 or Both)
E - WiFi (Cloud Connection)	J - Blower	O - AM or PM (Time)

Main Menu

Navigation

Navigating the entire menu structure is done with 2 or 3 buttons on the control panel.



Some panels have separate WARM (Up) and COOL (Down) buttons, while others have a single Temperature button. In the navigation diagrams Temperature buttons are indicated by a single button icon. Panels that have two Temperature buttons (Warm and Cool) can use both of them to simplify navigation and programming where a single Temperature icon is shown.

The MENU/SELECT Button is used to choose the various menus and navigate each section.

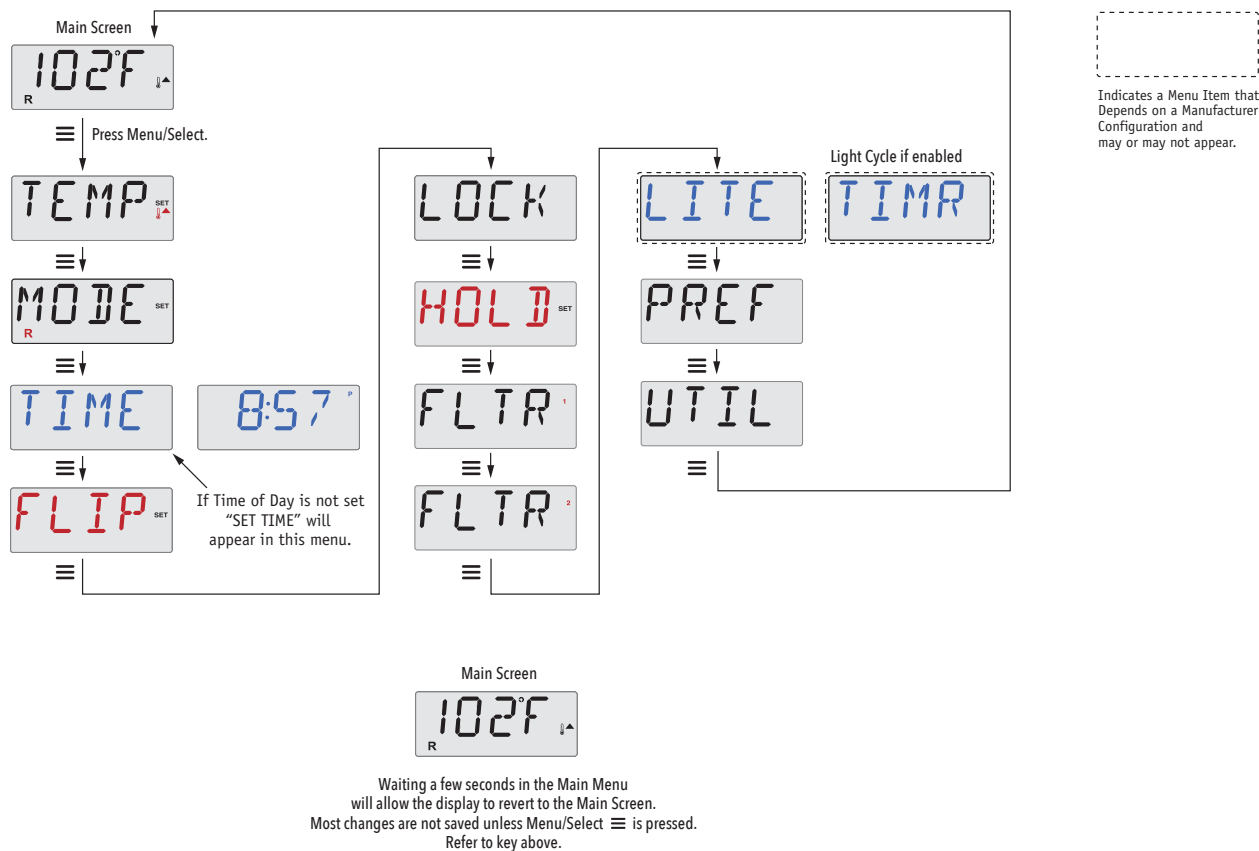
Typical use of the Temperature button(s) allows changing the Set Temperature while the numbers are flashing in the LCD. The menus can be exited with certain button presses. Simply waiting for a few seconds will return the panel operation to normal.

Power-up Screens

Each time the System powers up, a series of numbers is displayed. After the startup sequence of numbers, the system will enter Priming Mode (See Page 3).

Key

- Indicates Flashing or Changing Segment
- Indicates Alternating or Progressive Message - every 1/2 second
-  A temperature button, used for "Action"
-  Menu/Select button
- Waiting time that keeps the last change to a menu item.
- ***** Waiting time (depends on menu item) that reverts to original setting and ignores any change to that menu item.



Fill it up!

Preparation and Filling

Fill the spa to its correct operating level. Be sure to open all valves and jets in the plumbing system before filling to allow as much air as possible to escape from the plumbing and the control system during the filling process.

After turning the power on at the main power panel, the top-side panel display will go through specific sequences. These sequences are normal and display a variety of information regarding the configuration of the hot tub control.

Priming Mode - MO19*

This mode will last for 4-5 minutes or you can manually exit the priming mode after the pump(s) have primed.



Regardless of whether the priming mode ends automatically or you manually exit the priming mode, the system will automatically start normal heating and filtering at the end of the priming mode. During the priming mode, the heater is disabled to allow the priming process to be completed without the possibility of energizing the heater under low-flow or no-flow conditions. Nothing comes on automatically, but the pump(s) can be energized by pushing the “Jets” or “Aux” buttons.

If the spa has a Circ Pump, it can be activated by pressing the “Light” button during Priming Mode.

Priming the Pumps

As soon as the above display appears on the panel, push the “Jets” button once to start Pump 1 in low-speed and then again to switch to high-speed. Also, push the “Jets 2” or “Aux” button, if you have a 2nd pump, to turn it on. The pumps will now be running in high-speed to facilitate priming. If the pumps have not primed after 2 minutes, and water is not flowing from the jets in the spa, do not allow the pumps to continue to run. Turn off the pumps and repeat the process. Note: Turning the power off and back on again will initiate a new pump priming session. Sometimes momentarily turning the pump off and on will help it to prime. Do not do this more than 5 times. If the pump(s) will not prime, shut off the power to the spa and call for service.

Important: A pump should not be allowed to run without priming for more than 2 minutes. Under NO circumstances should a pump be allowed to run without priming beyond the end of the 4-5 minute priming mode. Doing so may cause damage to the pump and cause the system to energize the heater and go into an overheat condition.

Exiting Priming Mode

You can manually exit Priming Mode by pressing the “Warm” or “Cool” button. Note that if you do not manually exit the priming mode as described above, the priming mode will be automatically terminated after 4-5 minutes. Be sure that the pump(s) have been primed by this time.

Once the system has exited Priming Mode, the top-side panel will momentarily display the set temperature but the display will not show the water temperature yet, as shown below.

This is because the system requires approximately 1 minute of water flowing through the heater to determine the water temperature and display it.



*MO19 is a Message Code. See Page 18.

Spa Behavior

Pumps

Press the “Jets” button once to turn pump 1 on or off, and to shift between low and high speeds if equipped.

If left running, the pump will turn off after a time-out period.

On non-circ systems, the low-speed of pump 1 runs when the blower or any other pump is on. If the spa is in Ready Mode (See page 6), Pump 1 low may also activate once in a while for at least 1 minute to detect the spa temperature (polling) and then to heat to the set temperature if needed. When the low-speed turns on automatically, it cannot be deactivated from the panel, however the high speed may be started.

Circulation Pump Modes

If the system is equipped with a circ pump, it will be configured to work in one of three different ways:

- 1, The circ pump operates continuously (24 hours) with the exception of turning off for 30 minutes at a time when the water temperature reaches 3°F (1.5°C) above the set temperature (most likely to happen in very hot climates).
- 2, The circ pump stays on continuously, regardless of water temperature.
- 3, A programmable circ pump will come on when the system is checking temperature (polling), during filter cycles, during freeze conditions, or when another pump or blower is on.

The specific Circulation Mode that is used has been determined by the Manufacturer and cannot be changed in the field.

Filtration and Ozone

On non-circ systems, Pump 1 low and the ozone generator will run during filtration. On circ systems, the ozone will run with the circ pump.

The system is factory-programmed with one filter cycle that will run in the evening (assuming the time-of-day is properly set) when energy rates are often lower. The filter time and duration are programmable. (See page 10)

A second filter cycle can be enabled as needed.

At the start of each filter cycle, all water devices (other than the primary pump) will run briefly to purge the plumbing to maintain good water quality. The term “water devices” includes the Blower.

Freeze Protection

If the temperature sensors within the heater detect a low enough temperature, then the pump(s) and the blower automatically activate to provide freeze protection. The pump(s) and blower will run either continuously or periodically depending on conditions.

In colder climates, an optional freeze sensor may be added to protect against freeze conditions that may not be sensed by the standard sensors. Auxiliary freeze sensor protection acts similarly except with the temperature thresholds determined by the switch. See your dealer for details.

Clean-up Cycle (optional)

When a pump or blower is turned on by a button press, a clean-up cycle begins 30 minutes after the pump or blower is turned off or times out. The pump and the ozone generator will run for 30 minutes or more, depending on the system. On some systems, you can change this setting. (See the Preferences section on page 12)

Temperature and Temp Range

Adjusting the Set Temperature

When using a panel with Up and Down buttons (Temperature buttons), pressing Up or Down will cause the temperature to flash. Pressing a temperature button again will adjust the set temperature in the direction indicated on the button. When the LCD stops flashing, the spa will heat to the new set temperature when required.

If the panel has a single temperature button, pressing the button will cause the temperature to flash. Pressing the button again will cause the temperature to change in one direction (e.g. UP). After allowing the display to stop flashing, pressing the Temperature Button will cause the temperature to flash and the next press will change the temperature in the opposite direction (e.g. DOWN).

Press-and-Hold

If a Temperature button is pressed and held when the temperature is flashing, the temperature will continue to change until the button is released. If only one temperature button is available and the limit of the Temperature Range is reached when the button is being held, the progression will reverse direction.

Dual Temperature Ranges

This system incorporates two temperature range settings with independent set temperatures. The High Range designated in the display by a thermometer and an “up” arrow, and the Low Range designated in the display by a thermometer and “down” arrow.

These ranges can be used for various reasons, with a common use being a “ready to use” setting vs. a “vacation” setting. The Ranges are chosen using the menu structure below. Each range maintains its own set temperature as programmed by the user. This way, when a range is chosen, the spa will heat to the set temperature associated with that range.

For example:

High Range might be set between 80°F and 104°F.

Low Range might be set between 50°F and 99°F.

More specific Temp Ranges may be determined by the Manufacturer.

Freeze Protection is active in either range.

See Ready and Rest on Page 6

for additional heating control information.

Key

■ Indicates Flashing or Changing Segment

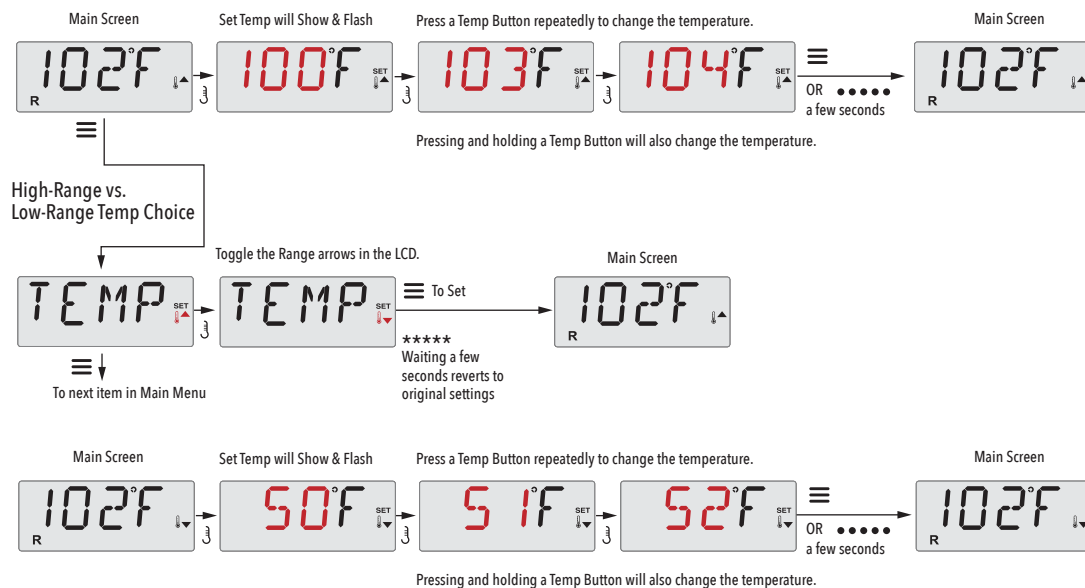
■ Indicates Alternating or Progressive Message - every 1/2 second

} A temperature button, used for “Action”

≡ Menu/Select button

••••• Waiting time that keeps the last change to a menu item.

***** Waiting time (depends on menu item) that reverts to original setting and ignores any change to that menu item.



Mode – Ready and Rest

In order for the spa to heat, a pump needs to circulate water through the heater. The pump that performs this function is known as the “primary pump.”

The primary pump can be either a 2-Speed Pump 1 or a circulation pump.

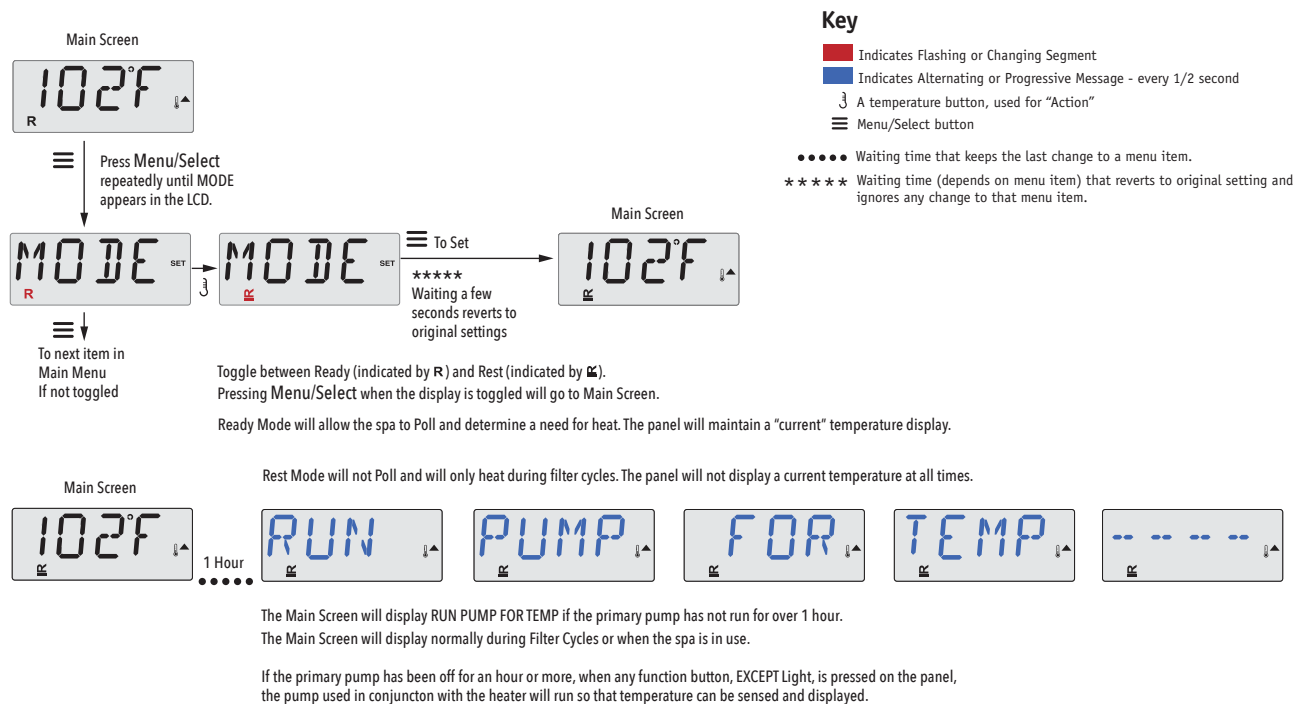
If the primary pump is a 2-Speed Pump 1, Ready Mode (indicated by **R**) will circulate water periodically, using Pump 1 Low, in order to maintain a constant water temperature, heat as needed, and refresh the temperature display. This is known as “polling.”

Rest Mode (indicated by **⏸**) will only allow heating during programmed filter cycles. Since polling does not occur, the temperature display may not show a current temperature until the primary pump has been running for a minute or two.

Circulation Mode (See Page 4, under Pumps, for other circulation modes)

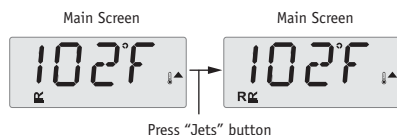
If the spa is configured for 24HR circulation, the primary pump generally runs continuously. Since the primary pump is always running, the spa will maintain set temperature and heat as needed in Ready Mode, without polling.

In Rest Mode, the spa will only heat to set temperature during programmed filter times, even though the water is being filtered constantly when in Circulation Mode.



Ready-in-Rest Mode

R ⏸ appears in the display if the spa is in Rest Mode and “Jets” is pressed. It is assumed that the spa is being used and will heat to set temperature. The primary pump will run until set temperature is reached, or 1 hour has passed. After 1 hour, the System will revert to Rest Mode. This mode can also be reset by entering the Mode Menu and changing the Mode.



Restricting Operation

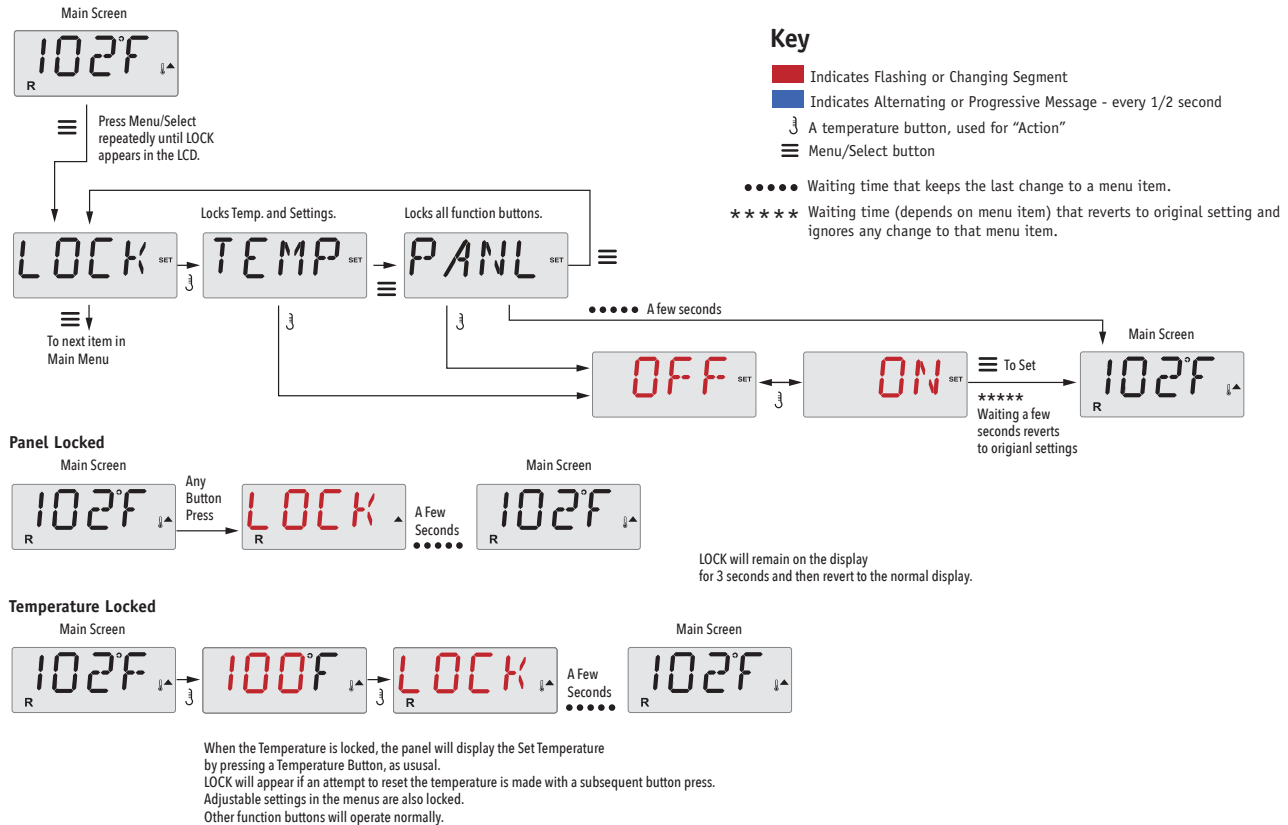
The control can be restricted to prevent unwanted use or temperature adjustments.

Locking the panel prevents the controller from being used, but all automatic functions are still active.

Locking the Temperature allows Jets and other features to be used, but the Set Temperature and other programmed settings cannot be adjusted.

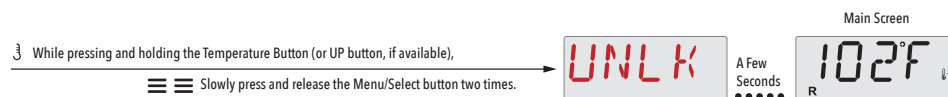
Temperature Lock allows access to a reduced selection of menu items.

These include Set Temperature, FLIP, LOCK, UTIL, INFO and FALT LOG.



Unlocking

This Unlock sequence may be used from any screen that may be displayed on a restricted panel.



NOTE: If the panel has both an UP and a Down button, the ONLY button that will work in the Unlock Sequence is the UP button.

The temperature will not Unlock if the Unlock sequence is done while the panel is displaying "LOCK".

Hold (Standby)

Hold Mode –MO37*

Hold Mode is used to disable the pumps during service functions like cleaning or replacing the filter. Hold Mode will last for 1 hour unless the mode is exited manually.

Drain Mode

Some spas have a special feature that allows a pump to be employed when draining the water.

When available, this feature is a component of Hold Mode.

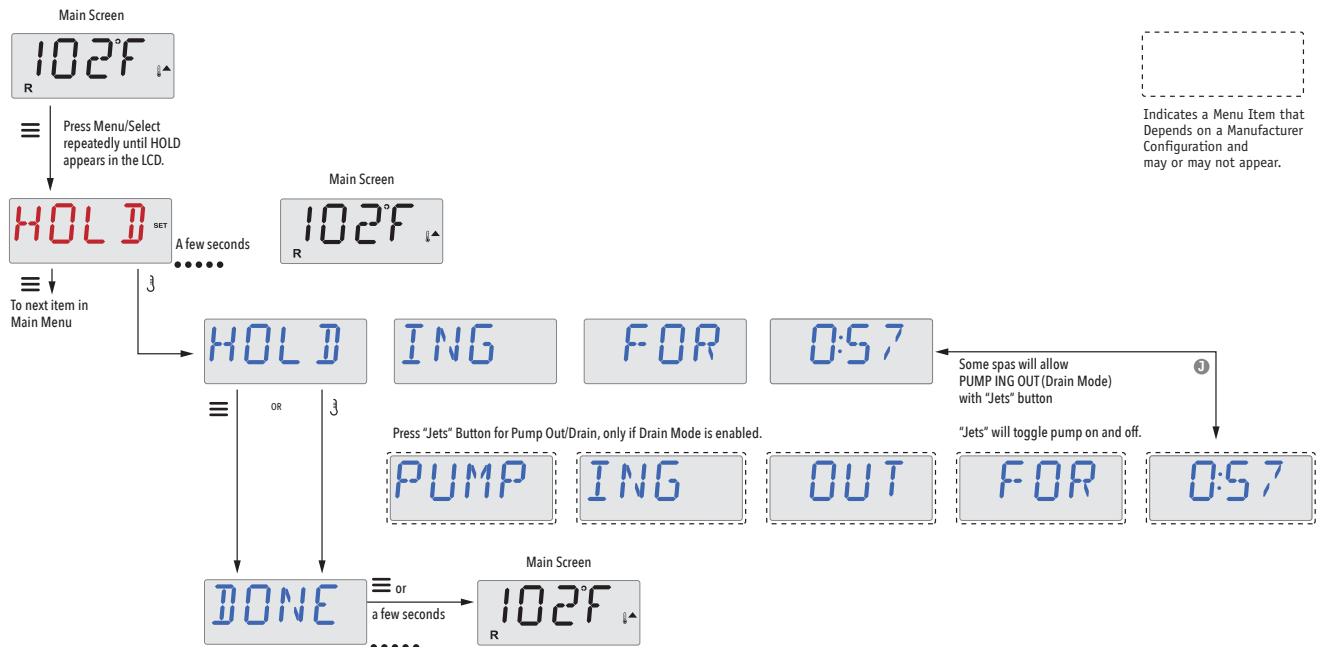
Drain Mode will time out with Hold Mode.

Key

- Indicates Flashing or Changing Segment
- Indicates Alternating or Progressive Message - every 1/2 second
- A temperature button, used for "Action"
- Menu/Select button

..... Waiting time that keeps the last change to a menu item.

***** Waiting time (depends on menu item) that reverts to original setting and ignores any change to that menu item.



Preferences

F / C (Temp Display)

Change the temperature between Fahrenheit and Celsius.

12 / 24 (Time Display)

Change the clock between 12 hr and 24 hr display.

RE-MIN-DERS (Reminders)

Turn the display of reminder messages (like "Clean Filter") On or Off.

Note: Reminders continue to run in the background even when not displayed. So turning the display of Reminders On or Off does not reset any Reminder counts.

CLN-UP (Cleanup)

Cleanup Cycle Duration is not always enabled, so it may not appear. When it is available, set the length of time Pump 1 will run after each use. 0-4 hours are available.

M8

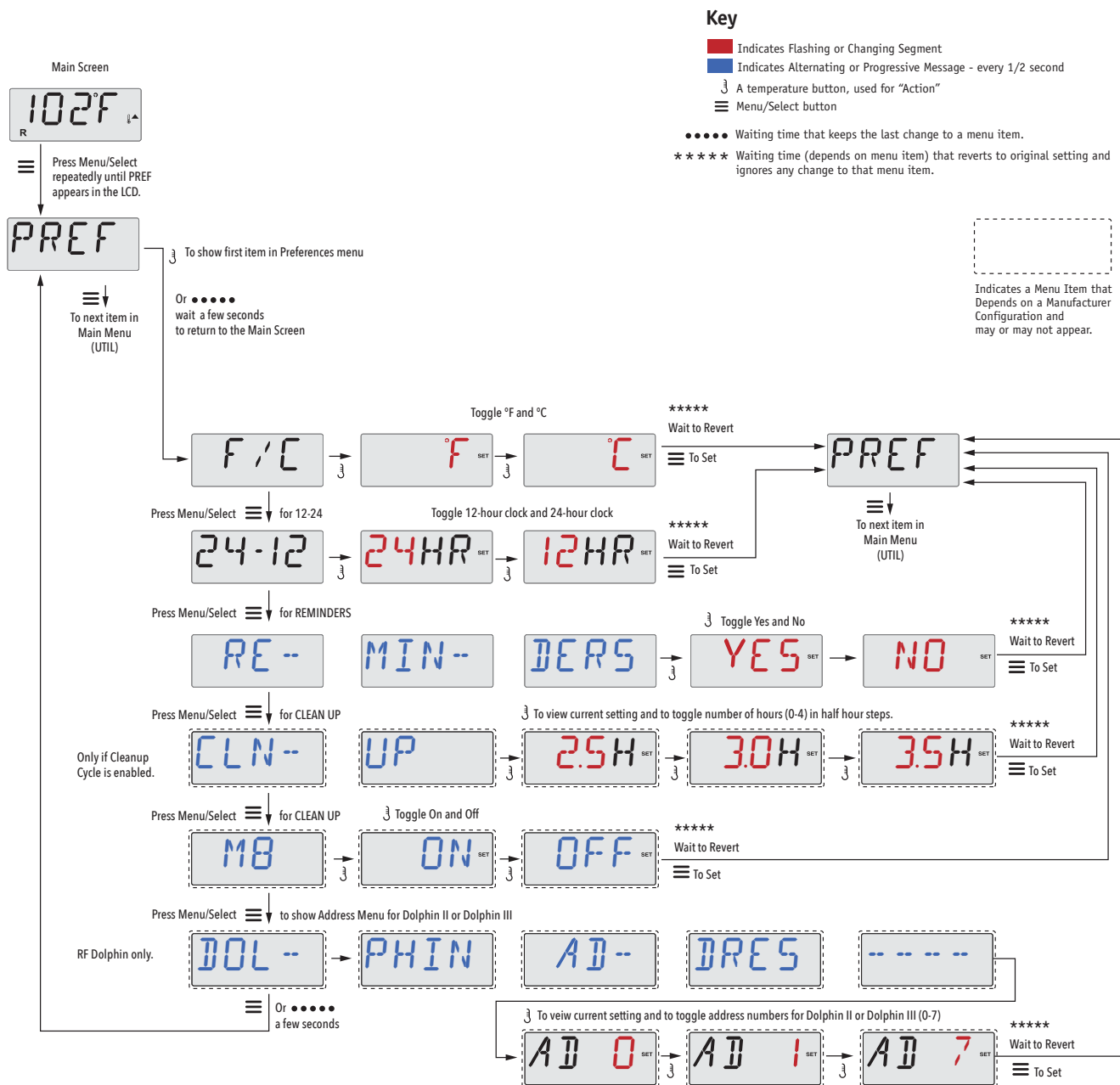
(This message may not appear on all systems.) On systems that have M8, it is enabled by default. It can be disabled (or re-enabled) here. M8 reduces polling intervals when the water temperature in the spa is steady.

DOL-PHIN-AD-DRES (Dolphin II and Dolphin III) Applies to RF Dolphin only.

(This message may not appear depending on the configuration)

When set to 0, no addressing is used. Use this setting for a Dolphin Remote which is factory set for no address by default. When set between 1 and 7, the number is the address. (See the Dolphin manual for details.)

Preferences



Utilities and Information

INFO (System Information sub-menu)

The System Information Menu displays various settings and identification of the particular system.

SSID (Software ID)

Displays the software ID number for the System.

MODL (System Model)

Displays the Model Number of the System.

SETP (Current Setup)

Displays the currently selected Configuration Setup Number.

Heater Voltage (Feature not used on CE rated systems.)

Displays the operating voltage configured for the heater.

Heater Wattage as Configured in Software (CE Systems Only.)

Displays a heater kilowatt rating as programmed into the control system software (1-3 or 3-6).

H _ (Heater Type)

Displays a heater type ID number.

SW _ (Dip Switch Settings)

Displays a number that represents the DIP switch positions of S1 on the main circuit board.

PANL (Panel Version)

Displays a number of the software in the topside control panel.

Additional Utilities

Utilities

In addition to INFO, The Utilities Menu contains the following:

GFCI (GFCI Test)

(Feature not available on CE rated systems.)

GFCI Test is not always enabled, so it may not appear. This screen allows the GFCI to be tested manually from the panel and can be used to reset the automatic test feature. If the GFCI Test Feature is reset, the device will trip within 7 days. (See Page 17)

A / B (A/B Sensor Temperatures)

When this is set to On, the temperature display will alternate to display temperature from Sensor A and Sensor B in the heater.

FALT LOG (Fault Log)

The Fault Log is a record of the last 24 faults that can be reviewed by a service tech.

DEMO (Demo Mode)

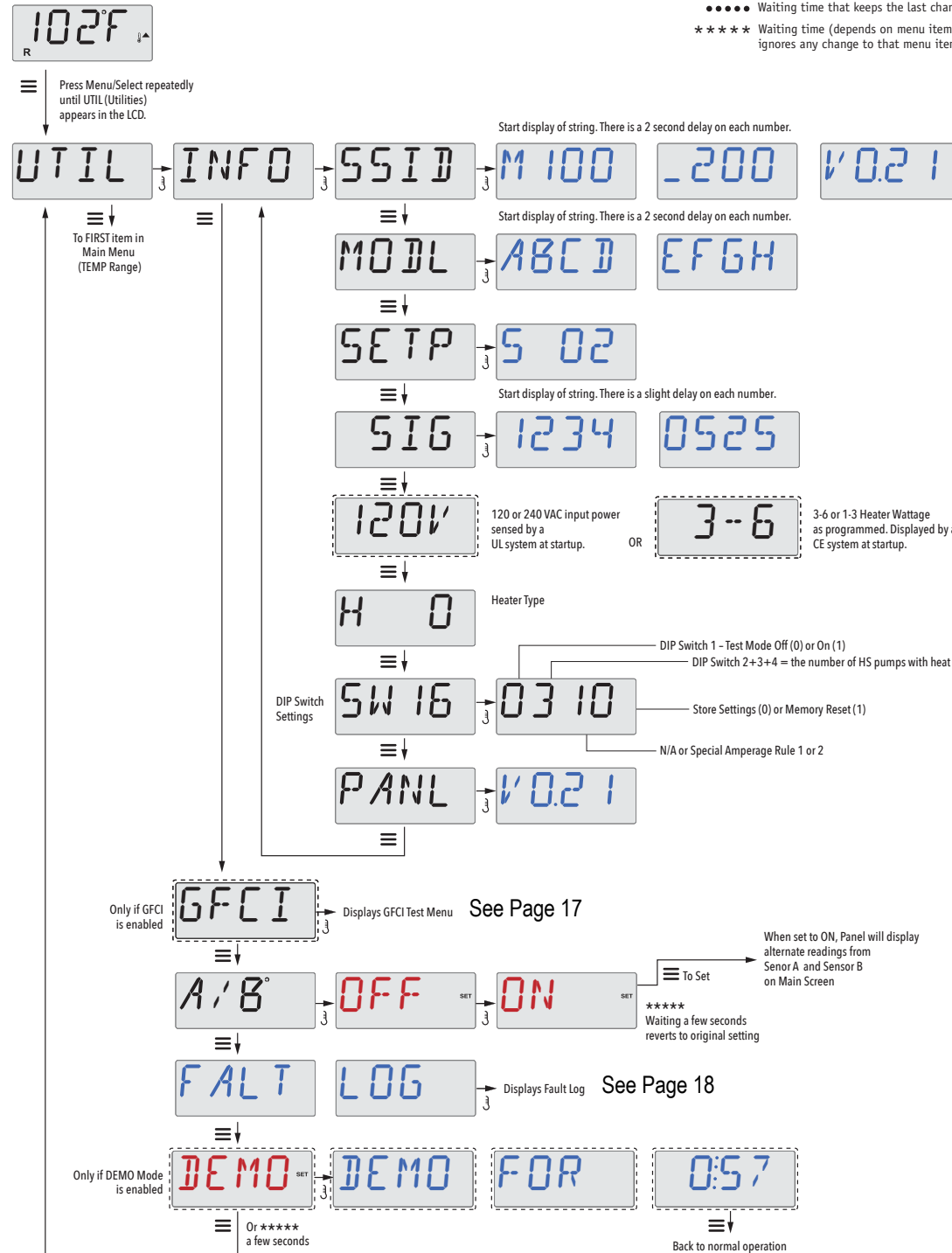
Demo Mode is not always enabled, so it may not appear. This is designed to operate several devices in a sequence in order to demonstrate the various features of a particular hot tub.

Utilities

Key

-  Indicates Flashing or Changing Segment
-  Indicates Alternating or Progressive Message - every 1/2 second
-  A temperature button, used for "Action"
-  Menu/Select button

- ● ● ● ● Waiting time that keeps the last change to a menu item.
- * * * * * Waiting time (depends on menu item) that reverts to original setting and ignores any change to that menu item.



These strings will display once and return to the beginning. Press Temperature again to display the string again.

●●●●●
Waiting 10 seconds will allow
the screen to return to
normal operation.

Indicates a Menu Item that Depends on a Manufacturer Configuration and may or may not appear.

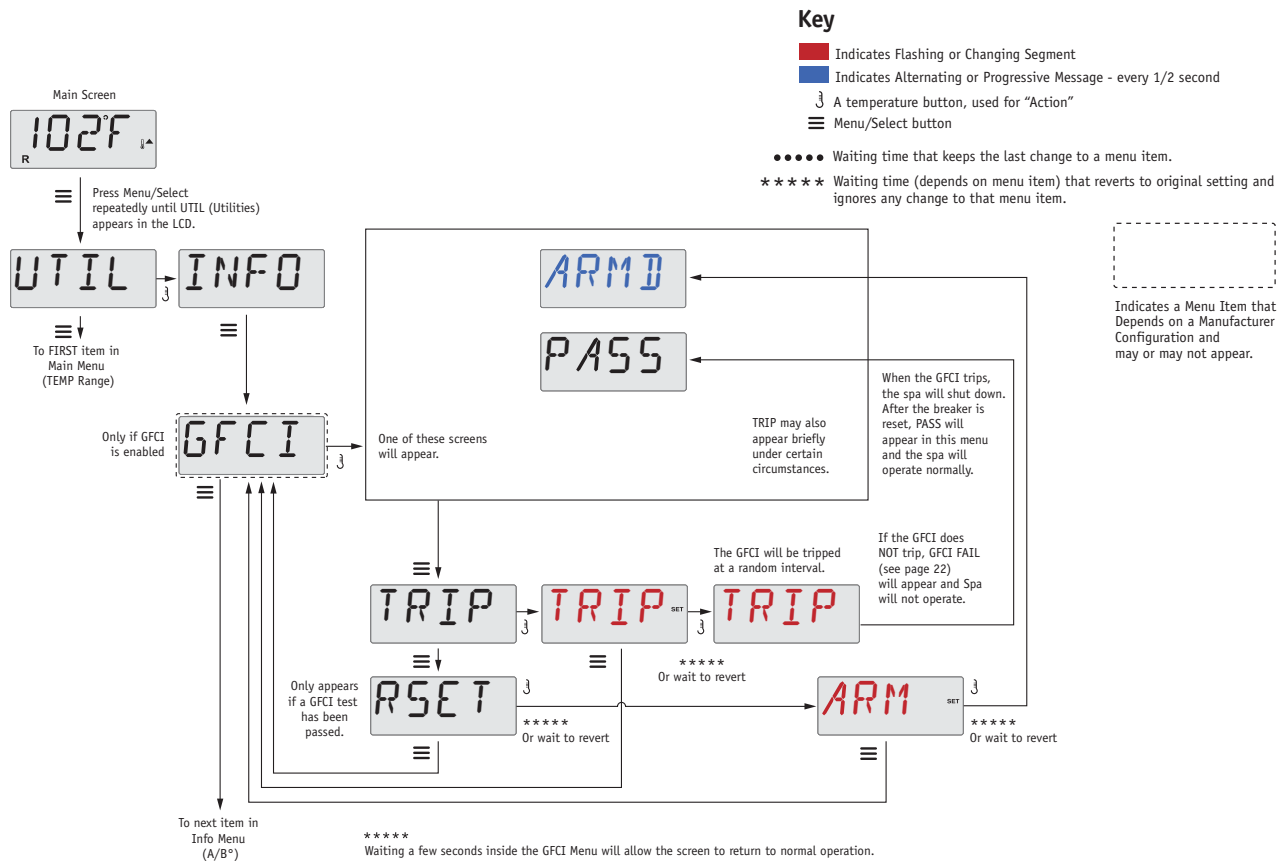
Utilities – GFCI Test Feature

Not Available on CE Rated Systems.

A GFCI is an important safety device and is required equipment on a hot tub installation.

Your spa may be equipped with a GFCI Protection feature. (UL rated systems only.) If your spa has this feature enabled by the manufacturer, the GFCI Trip Test must occur to allow proper spa function.

Within 1 to 7 days after startup, the spa will trip the GFCI to test it. (The number of days is factory programmed.) The GFCI must be reset once it has tripped. After passing the GFCI Trip Test, any subsequent GFCI trips will indicate a ground fault or other unsafe condition and the power to the spa must be shut off until a service person can correct the problem.



Forcing the GFCI Trip Test

The installer can cause the GFCI Trip Test to occur sooner by initiating it using the above menu.

The GFCI should trip within a few seconds and the spa should shut down. If it does not, shut down the power and manually verify that a GFCI breaker is installed and that the circuit and spa are wired correctly. Verify the function of the GFCI with its own test button. Restore power to the spa and repeat the GFCI Trip Test.

Once the GFCI is tripped by the test, reset the GFCI and the spa will operate normally from that point. You can verify a successful test by navigating to the above menu. PASS should appear after a temp button is pressed from the GFCI screen.

The end-user must be trained to expect this one-time test to occur and how to properly reset the GFCI.

Warning:

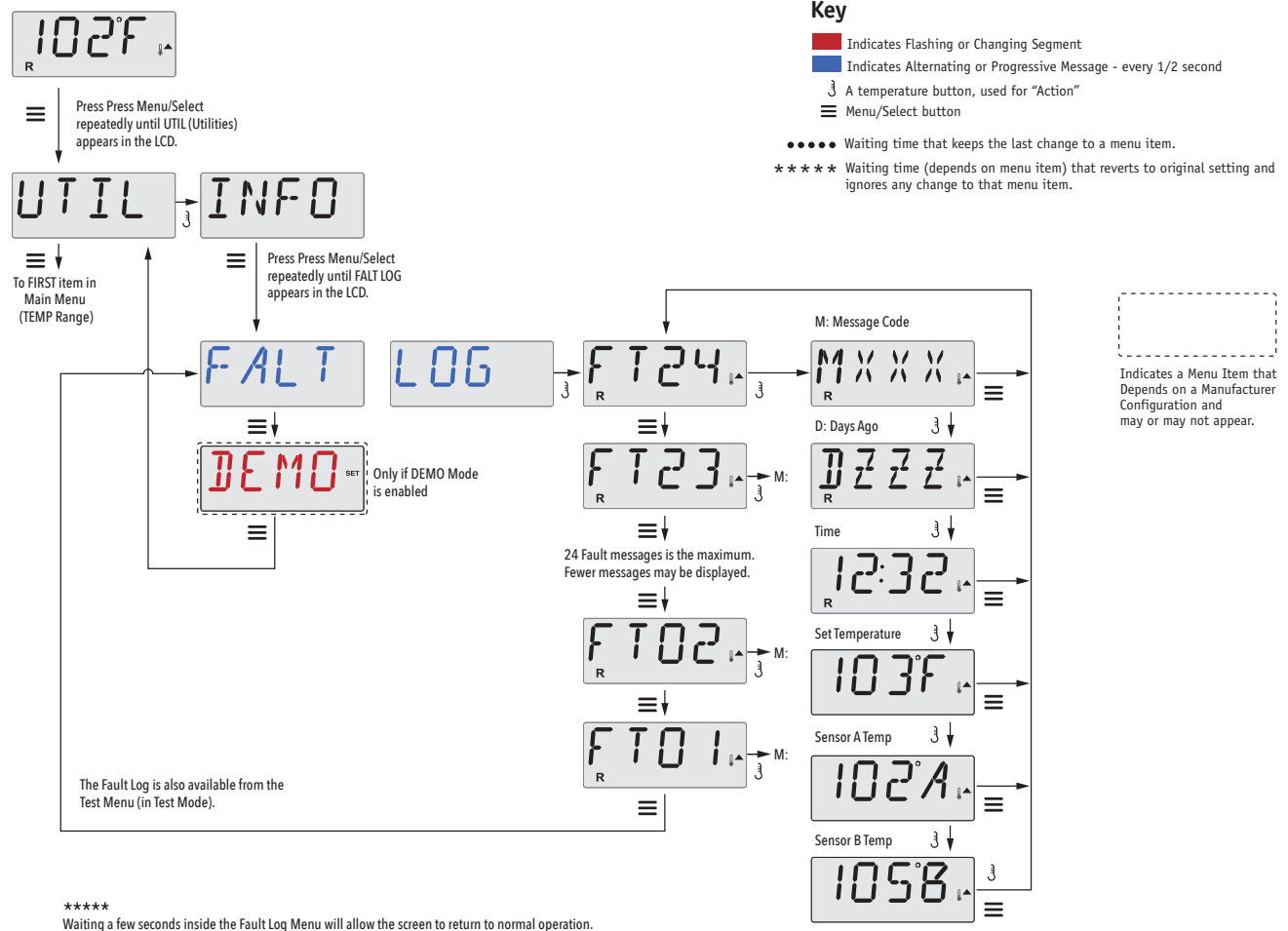
If freezing conditions exist, a GFCI should be reset immediately or spa damage could result. The end user should always be trained to test and reset the GFCI on a regular basis.

Utilities – Fault Log

A Little History can tell a lot

The Fault Log stores up to 24 events in memory and they can be reviewed under the Fault Log Menu.

Each event captures a Fault Message Code, how many days have passed since the fault, Time of the fault, Set Temperature during the fault, and Sensor A and B temperatures during the fault.



See following pages for various Message Codes and definitions.

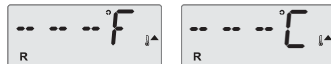
General Messages



Priming Mode – MO19

Each time the spa is powered up, it will enter Priming Mode. The purpose of Priming Mode is to allow the user to run each pump and manually verify that the pumps are primed (air is purged) and water is flowing. This typically requires observing the output of each pump separately, and is generally not possible in normal operation. Priming Mode lasts 4 minutes, but you can exit it earlier by pressing any Temp button. The heater is not allowed to run during Priming Mode.

NOTE: If your spa has a Circ Pump, it will turn on with “Light” in Priming Mode. The Circ Pump will run by itself when Priming Mode is exited.



Water Temperature is Unknown

After the pump has been running for 1 minute, the temperature will be displayed.



Too Cold - Freeze Protection

A potential freeze condition has been detected, or the Aux Freeze Switch has closed, and all pumps and blower are activated, either one at a time, or all at once, depending on how your system was built. All pumps and blower are ON for at least 4 minutes after the potential freeze condition has ended, or when the aux freeze switch opens.

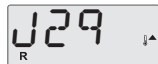
In some cases, pumps may turn on and off and the heater may operate during Freeze Protection.

This is an operational message, not an error indication.



Water is too Hot (OHS) – MO29

One of the water temp sensors has detected spa water temp 110°F (43.3°C) and spa functions are disabled. System will auto reset when the spa water temp is below 108°F (42.2°C). Check for extended pump operation or high ambient temp.



J29 Warning – MO44

J29 is typically used as a Heater Disable input. As such, it should not typically be shorted at power-up. This message appears if J29 is shorted at power-up.

MOXX numbers are Message Codes. See Page 18.

* This message can be reset from the topside panel with any button press.

Heater-Related Messages



Heater Flow is Reduced (HFL) – MO16

There may not be enough water flow through the heater to carry the heat away from the heating element. Heater start up will begin again after about 1 min. See “Flow Related Checks” below.



Heater Flow is Reduced (LF)* – MO17

There is not enough water flow through the heater to carry the heat away from the heating element and the heater has been disabled. See “Flow Related Checks” below. After the problem has been resolved, you must press any button to reset and begin heater start up.



Heater may be Dry (dr)* – MO28

Possible dry heater, or not enough water in the heater to start it. The spa is shut down for 15 min. Press any button to reset the heater start-up. See “Flow Related Checks” below.



Heater is Dry* – MO27

There is not enough water in the heater to start it. The spa is shut down. After the problem has been resolved, you must press any button to reset and restart heater start up. See “Flow Related Checks” below.



Heater is too Hot (OHH)* – MO30

One of the water temp sensors has detected 118°F (47.8°C) in the heater and the spa is shut down. You must press any button to reset when water is below 108°F (42.2°C). See “Flow Related Checks” below.



A Reset Message may Appear with other Messages.

Some errors may require power to be removed and restored.

Flow-Related Checks

Check for low water level, suction flow restrictions, closed valves, trapped air, too many closed jets and pump prime.

On some systems even when spa is shut down, some equipment may occasionally turn on to continue monitoring temperature or if freeze protection is needed.

* This message can be reset from the topside panel with any button press.

Sensor-Related Messages



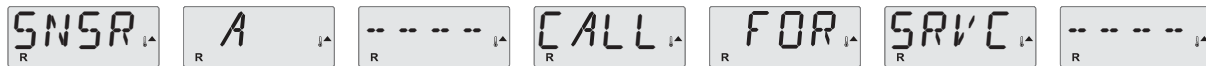
Sensor Balance is Poor – MO15

The temperature sensors MAY be out of sync by or 3°F. Call for Service.



Sensor Balance is Poor* – MO26

The temperature sensors ARE out of sync. The Sensor Balance is Poor fault has been established for at least 1 hour. Call for Service.



Sensor Failure – Sensor A: MO31, Sensor B: MO32

A temperature sensor or sensor circuit has failed. Call for Service.

Miscellaneous Messages



No Communications

The control panel is not receiving communication from the System. Call for Service.



Pre-Production Software

The Control System is operating with test software. Call for Service.

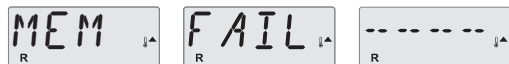


°F or °C is replaced by °T

The Control System is in Test Mode. Call for Service.

* This message can be reset from the topside panel with any button press.

System-Related Messages



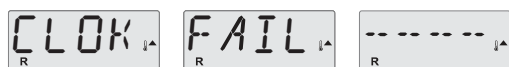
Memory Failure - Checksum Error* – MO22

At Power-Up, the system has failed the Program Checksum Test. This indicates a problem with the firmware (operation program) and requires a service call.



Memory Warning - Persistent Memory Reset* – MO21

Appears after any system setup change. Contact your dealer or service organization if this message appears on more than one power-up, or if it appears after the system has been running normally for a period of time.



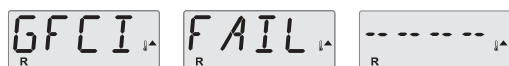
Memory Failure - Clock Error* – MO20 - Not Applicable on the BP1500

Contact your dealer or service organization.



Configuration Error – Spa will not Start Up

Contact your dealer or service organization.



GFCI Failure - System Could Not Test/Trip the GFCI – MO36

NORTH AMERICA ONLY. May indicate an unsafe installation. Contact your dealer or service organization.

* This message can be reset from the topside panel with any button press.

System-Related Messages



A Pump Appears to be Stuck ON – MO34

Water may be overheated. POWER DOWN THE SPA. DO NOT ENTER THE WATER. Contact your dealer or service organization.



A Pump Appears to have been Stuck ON when spa was last powered - MO35

POWER DOWN THE SPA. DO NOT ENTER THE WATER.

Contact your dealer or service organization.



The water level is too low

Some systems have a water level detect, and this message appears if it detects that the water level is too low.

* This message can be reset from the topside panel with any button press.

Reminder Messages

General maintenance helps.

The display of Reminder Messages can be suppressed by using the PREF Menu. See Page 12.

Reminder Messages can be chosen individually by the Manufacturer. They may be disabled entirely, or there may be a limited number of reminders on a specific model.

The frequency of each reminder (e.g. 7 days) can be specified by the Manufacturer.

Press a Temperature button to reset a displayed reminder message.



Alternates with temperature or normal display.

Appears on a regular schedule, e.g. every 7 days.

Check pH with a test kit and adjust pH with the appropriate chemicals.



Alternates with temperature or normal display.

Appears on a regular schedule, e.g. every 7 days.

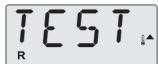
Check sanitizer level and other water chemistry with a test kit and adjust with the appropriate chemicals.



Alternates with temperature or normal display.

Appears on a regular schedule, e.g. every 30 days.

Clean the filter media as instructed by the manufacturer. See HOLD on page 9.



Alternates with temperature or normal display.

Appears on a regular schedule, e.g. every 30 days.

The Ground Fault Circuit Interrupter (GFCI) or Residual Current Device (RCD) is an important safety device and must be tested on a regular basis to verify its reliability.

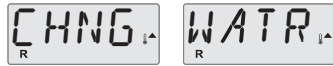
Every user should be trained to safely test the GFCI or RCD associated with the hot tub installation.

A GFCI or RCD will have a TEST and RESET button on it that allows a user to verify proper function.

Warning:

If freezing conditions exist, a GFCI or RCD should be reset immediately or spa damage could result. The end user should always be trained to test and reset the GFCI or RCD on a regular basis.

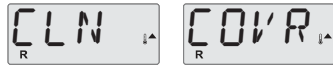
Reminder Messages Continued



Alternates with temperature or normal display.

Appears on a regular schedule, e.g. every 90 days.

Change the water in the spa on regular basis to maintain proper chemical balance and sanitary conditions.



Alternates with temperature or normal display.

Appears on a regular schedule, e.g. every 180 days.

Vinyl covers should be cleaned and conditioned for maximum life.



Alternates with temperature or normal display.

Appears on a regular schedule, e.g. every 180 days.

Wood skirting and furniture should be cleaned and conditioned per the manufacturers instructions for maximum life.



Alternates with temperature or normal display.

Appears on a regular schedule, e.g. every 365 days.

Filters should be replaced occasionally to maintain proper spa function and sanitary conditions.



Alternates with temperature or normal display.

As needed.

Install new mineral cartridge.



Alternates with temperature or normal display.

Appears on a regular schedule, e.g. every 365 days.

Check your ozone and/or UV generator per your spa manufacture's instructions.



Alternates with temperature or normal display.

Appears on a regular schedule, e.g. every 365 days.

Have a service technician do a check-up on your spa per your spa manufacturer's instructions.

Warning! Qualified Technician Required for Service and Installation

Basic Installation and Configuration Guidelines

Use minimum 6AWG copper conductors only.

Torque field connections between 21 and 23 in lbs.

Readily accessible disconnecting means to be provided at time of installation.

Permanently connected.

Connect only to a circuit protected by a Class A Ground Fault Circuit Interrupter (GFCI) or Residual Current Device (RCD) mounted at least 5' (1.52M) from the inside walls of the spa/hot tub and in line of sight from the equipment compartment.

CSA enclosure: Type 2

Refer to Wiring Diagram inside the cover of the control enclosure.

Refer to Installation and Safety Instructions provided by the spa manufacturer.

Warning: People with infectious diseases should not use a spa or hot tub.

Warning: To avoid injury, exercise care when entering or exiting the spa or hot tub.

Warning: Do not use a spa or hot tub immediately following strenuous exercise

Warning: Prolonged immersion in a spa or hot tub may be injurious to your health

Warning: Maintain water chemistry in accordance with the Manufacturers instructions.

Warning: The equipment and controls shall be located not less than 1.5 meters horizontally from the spa or hot tub.

Warning! GFCI or RCD Protection.

The Owner should test and reset the GFCI or RCD on a regular basis to verify its function.

Warning! Shock Hazard! No User Serviceable Parts.

Do not attempt service of this control system. Contact your dealer or service organization for assistance. Follow all owner's manual power connection instructions. Installation must be performed by a licensed electrician and all grounding connections must be properly installed.

CSA Compliance/Conformité

Caution:

- Test the ground fault circuit interrupter or residual current device before each use of the spa.
- Read the instruction manual.
- Adequate drainage must be provided if the equipment is to be installed in a pit.
- For use only within an enclosure rated CSA Enclosure 3.
- Connect only to a circuit protected by a Class A ground fault circuit interrupter or residual current device.
- To ensure continued protection against shock hazard, use only identical replacement parts when servicing.
- Install a suitably rated suction guard to match the maximum flow rate marked.

Warning:

- Water temperature in excess of 38°C may be injurious to your health.
- Disconnect the electrical power before servicing.

Attention:

- Toujours vérifier l'efficacité du disjoncteur différentiel avant d'utiliser différentiel avant d'utiliser le bain.
- Lire la notice technique.
- Lorsque l'appareillage est installé dans une fosse, on doit assurer un drainage adéquat.
- Employer uniquement à l'intérieur d'une clôture CSA Enclosure 3.
- Connecter uniquement à un circuit protégé par un disjoncteur différentiel de Class A.
- Afin d'assurer une protection permanente contre le danger de choc électrique, lors de l'entretien employer seulement des pièces de rechange identiques.
- Les prises d'aspiration doivent être équipées de grilles convenant au débit maximal indiqué.

Avertissement:

- Des températures de l'eau supérieures à 38°C peuvent présenter un danger pour la santé.
- Déconnecter du circuit d'alimentation électrique avant l'entretien.

Warning/Advertissement:

- Disconnect the electric power before servicing. Keep access door closed.
- Déconnecter du circuit d'alimentation électrique avant l'entretien. Garder la porte fermée.

TROUBLESHOOTING

DESCRIPTION OF TROUBLE	ANALYSIS	SOLUTION
An error message appears on the LCD display and the spa stops working	Error detected by digital diagnostics function	Please check the error message trouble shooting in the control systems manual, and then contact your local dealer if you are unable to resolve the issue
SPA can't start in any mode	- Power off - Control panel is locked - Control panel failure(display showed error message)	- Check the power switch on or leakage switch is turned on - Check the control panel is unlocked - Check the failure code in control system manual for solution then contact your dealer to do the maintenance
SPA starts automatically	- Filtration program is operating - Heating function is operating - Antifreezing function is operating	The control system will automatically stop when the program finishes
The water pump turns off	- The pump running time is more than the system default maximum running time - The pump motor is overheat protected	- Repress pump button to restart pump - Turn pump off for 1 Hour to cool and then start pump
Pump running after filling spa but no water coming through jets	- The jets are closed - Air blockage in the water pump	- The jets are closed - Turn pumps on and off several times to drain the blocked air. Unscrew the live union on the water pump to release the air and then screw on tightly. Note: Fill the spa with water through the filter to prevent air blockages in the pump (s)
Heater heating slowly or failing to heat	- The jets are closed - Air blockage in the water pump	- The jets are closed - Turn pumps on and off several times to drain the blocked air. Unscrew the live union on the water pump to release the air and then screw on tightly. Note: Fill the spa with water through the filter to prevent air blockages in the pump (s).
Reduced water flow through the jets	- Jets are not fully opened - Jets are blocked - Air valve is turned off - Water diverter closed to the seat	- Set the temperature up - Clean or replace paper filter core - Put the cover on the spa - Turn on the jets - Contact the local dealer for maintenance
Uneven water flow from jets	- Suctions or filter are blocked - Water pump side-check valves are not fully opened	- Clean the suction or filter - Check the water pump side-check valves are fully opened

THANK YOU
for choosing



